Exploiting SNOVA's Structure in the Wedge Product Attack

Maxime Bros[1,2], Thai Hung Le[3,4], Jacob Lichtinger[5], Brice Minaud[4], Ray Perlner[5], Daniel Smith-Tone[5,6], and Cristian Valenzuela[6]

[1] Associate, National Institute of Standards and Technology (NIST), Maryland, USA
maxime.bros@nist.gov
[2] Contractor, Izum Inc., Potomac, Maryland, USA
[3] LTCI, Telecom Paris, Institut Polytechnique de Paris, France
thai.le@telecom-paris.fr
[4] École normale supérieure, PSL University, CNRS, Inria, France
{hung.le,brice.minaud}@ens.fr
[5] National Institute of Standards and Technology (NIST), Maryland, USA
{jacob.lichtinger,ray.perlner,daniel.smith}@nist.gov
[6] University of Louisville, Louisville KY, USA
{dcsmit11,cristian.valenzuena}@louisville.edu

Abstract. Post-quantum cryptography (PQC) aims to develop cryptographic schemes secure against quantum adversaries. One promising class of digital signature schemes is based on multivariate quadratic equations, where Unbalanced Oil and Vinegar (UOV) is a leading example. UOV has been extensively studied since its introduction in 1999 [19], and it has remained secure. It offers very small signatures but suffers from very large public keys; to remediate this, some schemes—such as MAYO, QR-UOV, and SNOVA—add a structure to reduce the size of the public key. These four multivariate schemes are candidates that made it to the Second Round of the National Institute of Standards and Technology PQC Additional Call for Post-Quantum Signature schemes. In this work, we revisit a recently proposed algebraic attack by Ran [26,27] on UOV and extend this approach to a new attack on SNOVA by exploiting its block-ring structure. In addition to improving the attack complexity, our exploitation of the block-ring structure rules out spurious solutions, which prevents generic version of Ran's attack from applying to SNOVA. Our attack breaks 6 of the 11 currently proposed SNOVA parameter sets and improves on the previous best result for an additional 2 sets; it is significantly more effective against larger ℓ in comparison to several earlier attacks [5,9,17,20,22]. For example, for SNOVA-V with parameters $(v, o, \ell) = (29, 6, 5)$, the estimated security drops to 181 bits, compared to 310 bits for the previous best known attack.

Keywords: Post-Quantum Cryptography · UOV · SNOVA · Algebraic Attack.

1 Introduction

In 2016, the National Institute of Standards and Technology (NIST) began the process for standardizing quantum-resistant public-key cryptography. After selecting some algorithms to standardize, NIST opened an additional standardization process [23] to increase the variety of digital signature standards available. A significant number of submissions belong to the family of multivariate public-key schemes, with many based on the Unbalanced Oil and Vinegar (UOV) construction [19]. Among those which have advanced to the second round, only UOV-based schemes [6,7,15,30] have short signatures; however, another multivariate scheme MQOM [3] has also progressed, albeit following the MPC-in-the-Head paradigm and without short signatures.

Identifying schemes with short signatures was identified as a specific secondary goal in the NIST call for proposals [23]. However, most post-quantum candidates have significantly larger signatures than those of widely-used quantum-vulnerable schemes such as ECDSA. Among the candidates that advanced to the second round, only two broad families offer signature sizes of roughly 200 bytes or less: isogeny-based schemes and multivariate schemes. Isogeny-based schemes [1] are appealing for their combination of short signatures, compact public keys, and small expanded secret keys; however, they are significantly slower than UOV-based multivariate schemes, despite notable performance improvements in the second round. Moreover, their long-term security remains less understood, as highlighted by the recent attacks that broke SIDH and SIKE [10,21,28]. This uncertainty has motivated continued interest in fast and well-studied alternatives such as UOV-based designs within the multivariate family.

Unbalanced Oil and Vinegar. By comparison, UOV is a time-tested construction: despite extensive cryptanalytic efforts since its introduction in 1999, its security has remained robust. In addition to its long-standing security record, UOV offers excellent performance, with some of the fastest signing and verification speeds among post-quantum signature schemes.

UOV is a hash-and-sign multivariate signature scheme defined over a field $\mathbb{F}_q$ with $n = v + o$ variables and $m = o$ quadratic equations. Its trapdoor structure partitions the variables into *vinegar* (v) and *oil* (o) parts, with each public polynomial quadratic in all variables except that oil–oil terms are absent. Knowledge of the secret affine transformation T and the central map F enables the signer to efficiently invert the public map $P = F \circ T$. Without this extra information, the inversion reduces to solving a multivariate quadratic (MQ) system that is conjectured to be indistinguishable from random and computationally hard.

A significant drawback to the UOV scheme is its large public key size, which typically ranges in the dozens of kilobytes for compressed-key variants for NIST category 1 parameters. To address this issue, structured variants of UOV have been introduced, such as MAYO [6], QR-UOV [15], and SNOVA [30]. Similarly to structured lattices and codes, this additional structure allows for more efficient and compact schemes, but careful analysis is required to ensure that security is not weakened in the process. Previous notable attempts to reduce the public key size of UOV have included Rainbow [12], LUOV [8], and VOX [24]; however, these schemes all suffered severe attacks [4,13,14] during the NIST PQC Standardization process.

The SNOVA scheme. SNOVA is a round-2 candidate in the ongoing NIST standardization process for additional post-quantum signatures, noted for its significant public-key size reduction relative to UOV. For example, the smallest public-key size achieved by the submitted parameters of SNOVA is 1016 bytes (with a signature size of 248 bytes). SNOVA can be viewed as an augmentation of UOV with additional structure in two different ways: it embeds a MAYO-like "whipping" structure; and it uses a specific block-ring structure, similar to the one used in QR-UOV. Beullens showed that, in the initial design, the whipping component admits rank-based attacks when the emulsifier matrices allow low-rank linear combinations [5]. Independently, Cabarcas et al. and Furue et al. exploited the block-ring/group-action structure to accelerate polynomial-system solving for relevant SNOVA instances via extension-field changes of variables yielding ideals stable under a cyclic diagonal action and XL-style techniques leveraging multihomogeneity, leading to cheaper forgeries than earlier estimates in some settings [9,16]. Additional reductions relate SNOVA to UOV over modified parameters: Ikematsu et al. [17] and Li and Ding [20] convert SNOVA to UOV with parameters $(q, \ell v, \ell o)$, improving intersection attacks, while Nakamura et al. [22] reduces to a smaller UOV system over $\mathbb{F}_{q^\ell}$ with v vinegar and o oil variables, which for $\ell = 2$ fails to meet NIST security targets. Nevertheless, these attacks appear to become less effective as ℓ grows, while larger ℓ underlies SNOVA's advantage over other UOV-like constructions.

Our Contributions. In this work, we build on the wedge attack against generic UOV systems by Ran [26,27], which attacks UOV by linearly solving for an exterior space corresponding to the dual (annihilator) of the oil space. We identify and exploit the block-ring structure intrinsic to SNOVA's *mini-UOV* map (using the terminology of [2]), and this insight allows us to reduce the number of variables in the linear system attacking SNOVA from $\binom{\ell(v+o)}{\ell v}$ to $\binom{v+o}{v}^\ell$. Furthermore, this algebraic structure of SNOVA also enables us to derive an accurate formula for the rank of the linear system. In consequence, our attack breaks 6 of the 11 parameter sets and improves on the previous best results for an additional 2 sets. For example, for SNOVA-V with parameters $(v, o, \ell) = (29, 6, 5)$, our attack reduces the security to 181 bits, compared to 310 bits under the previous best known attack. The full implementation of our attack can be found at:

https://gitlab.di.ens.fr/hle/snova_wedge_attack

1.1 Notation

- Let q be a *fixed* power of 2 and $\mathbb{F}_q$ the finite field with q elements. In this article, $q = 16$. For $\ell \in \mathbb{N}$, $\mathbb{F}_{q^\ell}$ is its degree-ℓ extension. To denote an arbitrary field of characteristic 2, $\mathbb{F}$ is used.

- The set of matrices with a rows, b columns, and entries in $\mathbb{F}$ is denoted $\mathbb{F}^{a\times b}$.
- The general linear group of non singular matrices in $\mathbb{F}^{a\times a}$ is denoted $\mathrm{GL}_a(\mathbb{F})$.
- Matrices and vectors are written in boldface font: $\mathbf{M}$ and $\mathbf{v}$, respectively. Additionally, vectors are column vectors by default.
- The canonical basis for $\mathbb{F}^a$ is given by $\{\mathbf{e}_i\}_{i=1}^a$ where the i^{th} coordinate is 1 and the rest are 0.
- For some $k \in \mathbb{N}$, let $[k] = \{1, \dots, k\}$, and let Σ_k be the group of permutations on the set $[k]$.
- For $\mathbf{M} \in \mathbb{F}^{a\times b}$ and $J \subset [a]$ of size $|J| = k$, let $\mathbf{M}_{J*} \in \mathbb{F}^{k\times b}$ be the submatrix of $\mathbf{M}$ whose rows are indexed by J. Additionally, if $b \le a$ and $|J| = b$, $\det(\mathbf{M}_{J*})$ is a maximal minor of $\mathbf{M}$.
- UOV structure
  - Let $n, m \in \mathbb{N}$ and $\{p_i\}_{i=1}^m$ be quadratic maps on $\mathbb{F}^n$. Then for all $\mathbf{x} \in \mathbb{F}^n$, there are $\mathbf{P}_i \in \mathbb{F}^{n\times n}$ such that $p_i(\mathbf{x}) = \mathbf{x}^\intercal \mathbf{P}_i \mathbf{x}$.
  - In order to have a UOV structure, there exists an oil space $\mathcal{O} \subset \mathbb{F}^n$ of dimension $o = m$ such that for each $\mathbf{x} \in \mathcal{O}$ and $1 \le i \le m$, $p_i(\mathbf{x}) = 0$. Further, we can define the *annihilator space* $\mathcal{A} = \mathcal{O}^\perp$ as the left nullspace of $\mathcal{O}$, which has dimension $v = n - o$.
  - Without loss of generality and using the basis of the $\mathbf{P}_i$, there exists an $\mathbf{X} \in \mathbb{F}^{v\times o}$ such that bases of $\mathcal{O}$ and $\mathcal{A}$ are given by

$$\mathbf{O} = \begin{bmatrix} \mathbf{X} \\ \mathbf{I}_o \end{bmatrix} \text{ and } \mathbf{A} = \begin{bmatrix} \mathbf{I}_v \\ -\mathbf{X}^\intercal \end{bmatrix},$$

    respectively. Since we are in characteristic 2, we can omit the negative sign in the definition of $\mathbf{A}$. Then $\mathbf{A}^\intercal \mathbf{O} = 0$.
  - For $1 \le j \le v$, the j^{th} column of $\mathbf{A}$ is denoted $\mathbf{a}_j$.
- Tensor and wedge products
  - Let $\mathbf{B} \in \mathbb{F}^{a\times b}$ and $\mathbf{C} \in \mathbb{F}^{c\times d}$. Then the *tensor product* $\mathbf{B} \otimes \mathbf{C} \in \mathbb{F}^{ac\times bd}$ has its $(i,j)^{\text{th}}$ $c \times d$ block given by $\mathbf{B}_{i,j}\mathbf{C}$ for $1 \le i \le a$ and $1 \le j \le b$.
  - In particular, for vectors $\mathbf{x} \in \mathbb{F}^a$ and $\mathbf{y} \in \mathbb{F}^b$, $\mathbf{x} \otimes \mathbf{y}$ should be a vector of length ab. However, it is often more intuitive to think of $\mathbf{x} \otimes \mathbf{y}$ as $\mathbf{x}\mathbf{y}^\intercal \in \mathbb{F}^{a\times b}$.
  - For $k \in \mathbb{N}$, let $\{\mathbf{x}_i\}_{i=1}^k \subset \mathbb{F}^a$. Then the *wedge product* is denoted

$$\bigwedge_{i=1}^{k} \mathbf{x}_i = \mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_k := \sum_{\sigma \in \Sigma_k} \operatorname{sgn}(\sigma)\, \mathbf{x}_{\sigma(1)} \otimes \dots \otimes \mathbf{x}_{\sigma(k)}.$$

    Since $\mathbb{F}$ has characteristic 2, $\operatorname{sgn}(\sigma)$ can be omitted.
  - The linear span of all such wedge products of k vectors in $\mathbb{F}^a$ is denoted $\bigwedge^k \mathbb{F}^a$, whose elements are called *k-forms*. A *k-form* that can be decomposable into $\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_k$ is called a *k-blade*. Further, define $\bigwedge^0 \mathbb{F}^a = \mathbb{F}^a$ and $\bigwedge \mathbb{F}^a := \bigoplus_{k=0}^{a} \bigwedge^k \mathbb{F}^a$.

- An important property of the wedge product is if not all the $\mathbf{x}_i$ are unique, $\bigwedge_i \mathbf{x}_i = 0$.
- For the canonical basis $\{\mathbf{e}_i\}_{i=1}^a$ of $\mathbb{F}^a$ and $J \subset [a]$, $\mathbf{e}_J = \bigwedge_{j \in J} \mathbf{e}_j$.
- An *alternating bilinear form* can be represented by a matrix $\mathbf{B} \in \mathbb{F}^{a \times a}$ with 0 on the diagonal. As a result, $\mathbf{B}$ can also be represented as a 2-form:

$$\mathbf{B} = \sum_{1 \le i,j \le a} \mathbf{B}_{i,j} \mathbf{e}_i \mathbf{e}_j^\intercal = \sum_{1 \le i < j \le a} (\mathbf{B}_{i,j} + \mathbf{B}_{j,i})\, \mathbf{e}_i \wedge \mathbf{e}_j$$

- For a matrix (or vector) $\mathbf{B} \in \mathbb{F}^{a \times b}$, $\mathbf{B}^{\otimes c} = \mathbf{I}_c \otimes \mathbf{B}$ has c copies of $\mathbf{B}$ along the diagonal.

2 Structure of SNOVA's public key

In this section, we give the background information needed to understand our key recovery attack on SNOVA. SNOVA's public key consists of $m = o$ bilinear forms in $\ell n = \ell(v + o)$ variables, given by matrices $\mathbf{P}_i \in \mathbb{F}_q^{n\ell \times n\ell}$. These public matrices act on a pair of inputs $\mathbf{x}, \mathbf{y} \in \mathbb{F}_q^{n\ell}$ and vanish when both inputs lie in an secret subspace (the oil subspace) $\mathcal{O} \subset \mathbb{F}_q^{n\ell}$ of dimension $o\ell$. This oil subspace is structured so that it can be expressed as the column space of a matrix $\mathbf{O} \in \mathbb{F}_q^{n\ell \times o\ell}$, where each $\ell \times \ell$ block is restricted to be an element of the field $\mathbb{F}_q[\mathbf{S}]$, where $\mathbf{S}$ is a public symmetric $\ell \times \ell$ matrix with an irreducible characteristic polynomial over $\mathbb{F}_q$.

Each element of $\mathbb{F}_q[\mathbf{S}]$ has ℓ degrees of freedom, but a generic matrix in $\mathbb{F}_q^{\ell \times \ell}$ has ℓ^2. Since $\mathcal{O}$ lies over $\mathbb{F}_q[\mathbf{S}]$, so does its annihilator space $\mathcal{A}$. This extra structure creates some linear dependencies in the minor variables of $\mathcal{A}$, which are used to recover the oil space in this attack.

Finally, it should be noted that, while SNOVA's public key consists of only m bilinear forms, additional equations can be generated. As noted by [5] and several earlier attacks [17,20], an attacker can easily construct an $m\ell^2$-dimensional space of bilinear forms sharing the same oil space. This space is generated by the bilinear forms associated with matrices of the form $(\mathbf{S}^a)^{\otimes n} \mathbf{P}_\mathbf{i} (\mathbf{S}^b)^{\otimes n}$, for integers $0 \le a, b < \ell$. Just as the minor variables have extra linear dependencies, so do the $m\ell^2$ bilinear forms.

3 Wedge product attack

The public key of UOV consists of quadratic maps $p = (p_1, \ldots, p_m) : \mathbb{F}^n \to \mathbb{F}^m$ and the private key of a UOV instance is a secret subspace $\mathcal{O}$ on which the public key vanishes. An important map that can be associated to any quadratic map is its polar form.

Definition 1 (Polar form). *Let $p = (p_1, \ldots, p_m) : \mathbb{F}^n \to \mathbb{F}^m$ be a homogeneous quadratic map. The polar form Q of p is defined as*

$$\begin{aligned} &\mathbb{F}^n \times \mathbb{F}^n \to \mathbb{F}^m \\ &Q(\mathbf{x}, \mathbf{y}) \mapsto p(\mathbf{x} + \mathbf{y}) - p(\mathbf{x}) - p(\mathbf{y}) \end{aligned} \tag{1}$$

Note that the polar form is a symmetric bilinear map.

An alternative way of defining the polar form, where $p_i(\mathbf{x}) = \mathbf{x}^\intercal \mathbf{P}_i \mathbf{x}$, is to symmetrize the matrices $\mathbf{P}_i$. For the specific case of characteristic 2:

$$Q(\mathbf{x}, \mathbf{y}) := (Q_1, Q_2, \ldots, Q_m),$$

where each Q_i is defined as

$$Q_i(\mathbf{x}, \mathbf{y}) := \mathbf{x}^\intercal \mathbf{Q}_i \mathbf{y} \text{ where } \mathbf{Q}_i := \mathbf{P}_i + \mathbf{P}_i^\intercal. \tag{2}$$

Note that, for simplicity of exposition, the above definition addresses only the homogeneous case, relevant for any candidate scheme in the UOV subfamily. For the more general definition, applicable to any quadratic map, a constant correction term of $p(\mathbf{0})$ must be added to Equation (1) to achieve bilinearity.

An important observation is that the polar form is *an alternating bilinear form* in characteristic 2. Ran leveraged this fact and revealed that these public quadratics can be manipulated with exterior algebra to derive a new attack [26,27]. In order to describe the wedge product attack, we will describe some background information as appeared in [26].

Lemma 1 ([26], Lemma 2). *Let $\widehat{Q}_i \in \bigwedge^2 \mathbb{F}^n$ be an alternating form associated with the polar $Q_i(\mathbf{x}, \mathbf{y}) = 0$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{O}$, and let $\mathbf{a}_1, \ldots, \mathbf{a}_v$ be a basis of the annihilator space $\mathcal{A}$ of $\mathcal{O}$. Then there exists $\mathbf{x}_j \in \mathbb{F}^n$ such that*

$$\widehat{Q}_i = \sum_{j=1}^{v} \mathbf{a}_j \wedge \mathbf{x}_j$$

The next result is that there is a well-defined injection (also known as the *Plücker embedding*) that maps k-dimensional subspaces to (the projectivization of) a k-form. In particular,

$$\langle \mathbf{x}_1, \ldots, \mathbf{x}_k \rangle \mapsto \mathbf{x}_1 \wedge \cdots \wedge \mathbf{x}_k.$$

This will be used on the annihilator space $\mathcal{A}$ of $\mathcal{O}$, giving the following.

$$\mathcal{A} = \langle \mathbf{a}_1, \ldots, \mathbf{a}_v \rangle \mapsto \widehat{\mathcal{A}} := \mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_v.$$

Note that $\widehat{\mathcal{A}} \neq 0$ since the basis $\mathbf{A}$ is of full rank. Since each $\widehat{Q}_i$ has the form given above,

$$\widehat{\mathcal{A}} \wedge \widehat{Q}_i = \mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_v \wedge \sum_{j=1}^{v} \mathbf{a}_j \wedge \mathbf{x}_j = 0. \tag{3}$$

This property motivates the construction of the linear map

$$\begin{aligned} \Phi : \bigwedge^{v} \mathbb{F}^n &\to \left(\bigwedge^{v+2} \mathbb{F}^n \right)^m \\ \widehat{\mathcal{V}} &\mapsto (\, \widehat{\mathcal{V}} \wedge \widehat{Q}_1, \ldots, \widehat{\mathcal{V}} \wedge \widehat{Q}_m \,) \end{aligned} \tag{4}$$

for any v-form $\widehat{\mathcal{V}}$. If $\ker(\Phi)$ is one-dimensional, then its unique (up to scaling) kernel vector determines the annihilator space $\mathcal{A}$. Once $\mathcal{A}$ is known, the oil-vinegar decomposition can be recovered by applying Gaussian elimination (see [26], Section 3.3). In order to work with Φ, we can write the wedge products in terms of the maximal minors of any basis $\mathbf{A}$ of $\mathcal{A}$ using the following lemma.

Lemma 2 ([31], Proposition 3.3). *Let* $\mathbf{y}_1, \ldots, \mathbf{y}_k \in \mathbb{F}^a$ *and define the matrix*

$$\mathbf{Y} = \begin{bmatrix} \mathbf{y}_1 \cdots \mathbf{y}_k \end{bmatrix} = \begin{bmatrix} y_{1,1} & \cdots & y_{k,1} \\ \vdots & \ddots & \vdots \\ y_{1,a} & \cdots & y_{k,a} \end{bmatrix}$$

then

$$\mathbf{y}_1 \wedge \cdots \wedge \mathbf{y}_k = \sum_{\substack{J \subset [a] \\ |J|=k}} \det(\mathbf{Y}_{J*}) \cdot \mathbf{e}_J.$$

Using the canonical basis for $\mathbb{F}^n$ and the notation for generic UOV, we can write:

$$\widehat{Q}_i = \sum_{j<k \in [n]} \left(\mathbf{Q}_i\right)_{j,k} \mathbf{e}_j \wedge \mathbf{e}_k,$$

Because all coefficients $\left(\mathbf{Q}_i\right)_{j,k}$ are known, we can solve for $\widehat{\mathcal{A}}$ using Equation 3:

$$\begin{aligned} 0 &= \widehat{\mathcal{A}} \wedge \widehat{Q}_i \\ &= \sum_{j<k \in [n]} \left(\mathbf{Q}_i\right)_{j,k} \widehat{\mathcal{A}} \wedge \mathbf{e}_j \wedge \mathbf{e}_k \\ &= \sum_{j<k \in [n]} \left(\mathbf{Q}_i\right)_{j,k} \sum_{\substack{J \subset [n] \\ |J|=v}} \det(\mathbf{A}_{J*})\, \mathbf{e}_J \wedge \mathbf{e}_j \wedge \mathbf{e}_k \\ &= \sum_{\substack{K \subset [n] \\ |K|=v+2}} \left(\sum_{\substack{j<k \in K \\ J=K\setminus\{j,k\}}} \left(\mathbf{Q}_i\right)_{j,k} \det(\mathbf{A}_{J*}) \right) \mathbf{e}_K. \end{aligned}$$

Since $\{\mathbf{e}_K\}_{\substack{K\subset[n]\\|K|=v+2}}$ forms a basis of $\bigwedge^{v+2}\mathbb{F}^n$, this simplifies to:

$$\sum_{\substack{j<k\in K\\J=K\setminus\{j,k\}}}(\mathbf{Q}_i)_{j,k}\det(\mathbf{A}_{J*})=0. \tag{5}$$

Ranging over all $i\in[m]$, we have a linear system of $m\binom{n}{v+2}$ equations over $\binom{n}{v}$ variables, i.e., the maximal minors $\det(\mathbf{A}_{J*})$ of $\mathbf{A}$. However, not all of these equations are linearly independent and previous works [25,26,27] rely on the following conjecture.

Conjecture 1 ([27], Section 4.1). With high probability, the number of linear independent equations in such a linear system when $v<\min\{\frac{o-1}{2}\ ,\ 2\}m$ is

$$\operatorname{rank}\Phi=\min\left\{\binom{v+o}{v}-1\ ,\ \sum_{i=1}^{\lfloor\frac{o}{2}\rfloor}(-1)^{i+1}\binom{m+i-1}{i}\binom{v+o}{v+2i}\right\}. \tag{6}$$

Hence, the linear system is expected to be solvable with a unique solution for those (v,o,m)-UOV instances that satisfy

$$\sum_{i=0}^{\lfloor\frac{o}{2}\rfloor}(-1)^{i}\binom{m+i-1}{i}\binom{v+o}{v+2i}\ \le\ 1. \tag{7}$$

We discuss some conditions needed so that Conjecture 1 holds and some supporting arguments that the Conjecture holds under the UOV assumption in Appendix A.

Let E denote the number of linear equations and U the number of variables in the linear system encoded by Φ. Whenever condition (7) holds, the system induced by Φ is *overdetermined* in the sense that the rank estimate given by the second component of the min in Equation (6) is at least the number of unknowns $U=\binom{v+o}{v}$, so for a *generic* instance one would not expect any nonzero kernel vector to exist. In our setting, however, the public forms $\widehat{Q}_1,\dots,\widehat{Q}_m$ satisfy the UOV structure which forces the annihilator subspace $\mathcal{A}$ to lie in the kernel of Φ. Thus, condition (7) should be interpreted as follows: it is strong enough to make the system overdetermined, so we do not expect *additional* solutions beyond the one enforced by the UOV structure. In particular, we expect $\dim\ker(\Phi)=1$ so recovering the vinegar subspace reduces to extracting the unique kernel vector in $\ker(\Phi)$, corresponding to $\widehat{\mathcal{A}}$.

Eventually, we obtain a sparse matrix of size $E\times U$ whose row density is $\binom{v+2}{2}$, and whose rank is given by Conjecture 1. Using Wiedemann's algorithm for sparse linear algebra, the cost of finding a nonzero kernel vector is

$$\mathcal{O}\left(\binom{v+2}{2}E\,U\right)$$

field operations in $\mathbb{F}$. Therefore, the wedge-product viewpoint turns the algebraic cryptanalysis of UOV into a single structured linear problem that can be efficiently solved by modern sparse-matrix solvers.

4 Wedge product attack on SNOVA

In this section, we apply the wedge product attack to SNOVA and demonstrate that the scheme's specific algebraic structure can be exploited with great efficiency. Again, we consider the linear mapping Φ from Equation (4). When applied to SNOVA, an attacker may attempt to find a kernel vector $\widehat{\mathcal{A}} \in \ker(\Phi_{\mathrm{SNOVA}})$, where the oil and vinegar parameters are scaled by ℓ and the number of 2-forms by ℓ^2, as follows:

$$\begin{aligned} \Phi_{\mathrm{SNOVA}} : \bigwedge^{\ell v} \mathbb{F}_q^{\ell n} &\longrightarrow \left(\bigwedge^{\ell v+2} \mathbb{F}_q^{\ell n} \right)^{m\ell^2} \\ \widehat{\mathcal{V}} &\longmapsto (\, \widehat{\mathcal{V}} \wedge \widehat{Q}_1, \ldots, \widehat{\mathcal{V}} \wedge \widehat{Q}_{m\ell^2} \,) \end{aligned}$$

In the following sections, we study the exploitation of the block-ring structure of SNOVA to reduce both $\dim(\Phi_{\mathrm{SNOVA}})$ and $\mathrm{rank}(\Phi_{\mathrm{SNOVA}})$, thus reducing the computational cost of $\ker(\Phi_{\mathrm{SNOVA}})$.

4.1 Dimension of Φ_{SNOVA}

The variables in the wedge attack correspond to the maximal minors of an $\ell(v+o) \times \ell v$ matrix, denoted $\mathbf{A}$, representing the annihilator space $\mathcal{A}$ of $\mathcal{O}$. In the UOV setting, this matrix is generic; the dimension of Φ is $\binom{v+o}{v}$, which is the number of linearly independent maximal minors and thus the number of variables in the attack. However, there is an extra structure built into SNOVA: Let $\mathbf{S} \in \mathbb{F}_q^{\ell \times \ell}$ be a symmetric matrix with an irreducible characteristic polynomial. Then each $\ell \times \ell$ block of $\mathbf{A}$ is given by an element of $\mathbb{F}_q[\mathbf{S}]$, which is defined as:

$$\mathbb{F}_q[\mathbf{S}] = \{ b_0 + b_1\mathbf{S} + \cdots + b_{\ell-1}\mathbf{S}^{\ell-1} \mid b_i \in \mathbb{F}_q \} \cong \mathbb{F}_{q^\ell}.$$

This extra structure from the SNOVA construction changes the analysis of the resulting system.

Observe that this structure induces linear dependencies among the maximal minors, effectively reducing the number of variables in the system. In particular, for any $\mathbf{\Gamma} \in \mathbb{F}_q[\mathbf{S}]$, there is the relation:

$$\mathbf{\Gamma}^{\otimes(v+o)} \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{\Gamma}^{\otimes v}. \tag{8}$$

This identity can be used to express the minors of $\mathbf{A}$ given by $(\mathbf{e}_{i_1}^\intercal \cdot \mathbf{A}) \wedge \cdots \wedge (\mathbf{e}_{i_{v\ell}}^\intercal \cdot \mathbf{A})$ as a linear combination of other minors of $\mathbf{A}$, specifically:

$$(\mathbf{e}_{i_1}^\intercal \cdot \mathbf{A}) \wedge \cdots \wedge (\mathbf{e}_{i_{v\ell}}^\intercal \cdot \mathbf{A}) = \det(\mathbf{\Gamma})^{-v} \cdot (\mathbf{e}_{i_1}^\intercal \cdot \mathbf{\Gamma}^{\otimes(v+o)}\mathbf{A}) \wedge \cdots \wedge (\mathbf{e}_{i_{v\ell}}^\intercal \cdot \mathbf{\Gamma}^{\otimes(v+o)}\mathbf{A}).$$

One therefore expects that the number of linearly independent minors in SNOVA is less than $\binom{\ell(v+o)}{\ell v}$, the expected number of linearly independent minors for a generic UOV scheme with the same dimensions. We did some experiments with small parameters and found the number of linearly independent minors of SNOVA to be less than the total number of maximal minors. From these data, we hypothesized a formula for the number of linearly independent minors, see the third column in Table 1, which was then proved in Proposition 1.

Table 1. Number of linearly independent minors for small SNOVA instances

Parameters $(\boldsymbol{v}, \boldsymbol{o}, \boldsymbol{\ell})$	**# l.i. minors**	**# from Prop. 1**	**Total # max minors**
$(2,2,2)$	36	36	70
$(2,1,3)$	27	27	84
$(2,2,3)$	216	216	924
$(3,1,2)$	16	16	28
$(2,1,4)$	81	81	495

Proposition 1. *Let $\iota : \mathbb{F}_q[\mathbf{S}] \hookrightarrow \mathbb{F}_q^{\ell\times\ell}$ be the canonical injection. Given this injection, a matrix $\mathbf{M} \in (\mathbb{F}_q[\mathbf{S}])^{a\times b}$ induces a matrix $\iota(\mathbf{M}) \in \mathbb{F}_q^{a\ell\times b\ell}$. By construction, the matrix $\mathbf{A}$ corresponding to the annihilator space is an element of $(\mathbb{F}_q[\mathbf{S}])^{n\times v}$, so $\iota(\mathbf{A}) \in \mathbb{F}_q^{n\ell\times v\ell}$. Then the number of linearly independent maximal minors of $\iota(A)$ is*

$$\binom{n}{v}^{\ell}.$$

Proof. Recall that a square matrix over a finite field with an irreducible characteristic polynomial is diagonalizable over the splitting field of the characteristic polynomial [22]. Thus, for the matrix $\mathbf{S} \in \mathbb{F}_q^{\ell\times\ell}$ defined in the specification [30], there exists a matrix $\mathbf{B} \in \mathrm{GL}_\ell(\mathbb{F}_{q^\ell})$ such that $\mathbf{B}^{-1}\mathbf{S}\mathbf{B} \in \mathbb{F}_{q^\ell}^{\ell\times\ell}$ is a diagonal matrix. In particular, all elements of $\mathbb{F}_q[\mathbf{S}]$ are simultaneously diagonalizable with this matrix $\mathbf{B}$.

For any $a \in \mathbb{N}$, let $\mathbf{U}_{a\ell}$ be the diagonal block matrix with a copies of the matrix $\mathbf{B}$, namely,

$$\mathbf{U}_{a\ell} = \mathbf{I}_a \otimes \mathbf{B} \in \left(\mathbb{F}_{q^\ell}\right)^{a\ell\times a\ell}.$$

The block components of the matrix $\mathbf{A}$, which is an element of $(\mathbb{F}_q[\mathbf{S}])^{n\times v}$ are diagonalized with $\mathbf{U}_{n\ell}$ and $\mathbf{U}_{v\ell}$ as $\mathbf{A}_1 = \mathbf{U}_{n\ell}^{-1}\,\iota(\mathbf{A})\,\mathbf{U}_{v\ell}$. This operation transforms the whole matrix into block form, where each block is a diagonal matrix.

For any matrix $\mathbf{V} \in \mathbb{F}_q^{n\ell\times v\ell}$, let $\det(\mathbf{V}_{J*})$ denote the maximal minors of $\mathbf{V}$ indexed by row subsets $J \subset [n\ell]$ of size $|J| = v\ell$, and define the vector of minors

$$\Delta(\mathbf{V}) := (\det(\mathbf{V}_{J*}))_{\substack{J\subset[n\ell]\\|J|=v\ell}} \in \mathbb{F}_q^N, \text{ with } N = \binom{n\ell}{v\ell}.$$

We will abuse notation and use $\det(\mathbf{A}_{J*})$ and $\Delta(\mathbf{A})$ for the minors of $\iota(\mathbf{A})$. Multiplying on the right by $\mathbf{U}_{v\ell}$ gives

$$\det(\iota(\mathbf{A})\,\mathbf{U}_{v\ell})_{J*} = \det(\mathbf{A}_{J*}) \cdot \det(\mathbf{U}_{v\ell}) \text{ for all } J,$$

so all minors are rescaled by the same nonzero constant $\det(\mathbf{U}_{v\ell}) \in \mathbb{F}_q^{\times}$.

Left multiplication changes the rows: if $\mathbf{x}_1, \ldots, \mathbf{x}_{n\ell}$ are the rows of $\mathbf{A}$, then the rows of $\mathbf{A}_1$ are $\mathbf{y}_j = \sum_{k=1}^{n\ell} (\mathbf{U}_{n\ell}^{-1})_{j,k}\mathbf{x}_k$. Applying the Cauchy-Binet formula yields:

$$\det(\mathbf{U}_{n\ell}^{-1}\,\iota(\mathbf{A}))_{J*} = \det\big((\mathbf{U}_{n\ell}^{-1})_{J*}\,\iota(\mathbf{A})\big) = \sum_{\substack{I\subset[n\ell]\\|I|=v\ell}} \det\big((\mathbf{U}_{n\ell}^{-1})_{J,I}\big) \cdot \det(\mathbf{A}_{I*})$$

This transformation is linear and invertible because $\mathbf{U}_{n\ell}$ is invertible, so the coefficient matrix $\mathbf{C} = \big(\det((\mathbf{U}_{n\ell})_{J,I})\big)_{J,I}$ is also invertible. Thus,

$$\Delta(\mathbf{A}_1) = \det(\mathbf{U}_{v\ell}) \cdot \mathbf{C}\big(\Delta(\mathbf{A})\big).$$

Since $\mathbf{C}$ is invertible, the linear span of the minors remains unchanged. Therefore, diagonalizing $\mathbf{A}$ over the extension field does *not* change the number of linearly independent maximal minors over the base field.

Because every $\ell \times \ell$ block in $\mathbf{A}_1$ is itself diagonal, the matrix $\mathbf{A}_1$ is block-diagonal when viewed in a different basis. In other words, there exist two permutation matrices $\mathbf{\Pi}_{n\ell}$ and $\mathbf{\Pi}_{v\ell}$ such that

$$\mathbf{A}_2 = \mathbf{\Pi}_{n\ell}\mathbf{A}_1\mathbf{\Pi}_{v\ell} = \operatorname{diag}(\mathbf{B}_1, \mathbf{B}_2, ..., \mathbf{B}_\ell).$$

Since $\mathbf{\Pi}_{n\ell}$ and $\mathbf{\Pi}_{v\ell}$ are permutation matrices, they only reorder the rows and columns of $\mathbf{A}_1$ and do not introduce any new linear dependencies among its maximal minors. As a result, the number of linearly independent maximal minors of $\mathbf{A}_2$ remains the same in $\mathbf{A}_1$.

In the matrix $\mathbf{A}_2$, each block $\mathbf{B}_i$ contributes to the maximal minors. To form a non-zero maximal minor, we must select exactly v rows from the n available rows in each $\mathbf{B}_i$; selecting fewer or more would result in a minor with zero determinant due to the structure of $\mathbf{A}_2$.

Moreover, because each $\mathbf{B}_i$ occupies a distinct set of columns in $\mathbf{A}_2$, the choices of rows in different blocks are independent of each other. Therefore, the total number of linearly independent maximal minors in $\mathbf{A}_2$ is:

$$\binom{n}{v}^{\ell}.$$

□

When applying the results from Proposition 1 to the parameters of SNOVA, it saves about $2 - 9$ bits in variable count (see Table 2).

Table 2. Comparison of the number (given as a logarithm in base 2) of linearly independent (l.i.) minors vs. total number of minors across SNOVA instances.

Variants	v	o	q	ℓ	# l.i. minors	# minors
SNOVA-I	37	17	16	2	91	93
	25	8	16	3	71	76
	24	5	16	4	67	74
SNOVA-III	56	25	16	2	138	141
	49	11	16	2	115	120
	37	8	16	4	111	118
	24	5	16	5	85	95
SNOVA-V	75	33	16	2	185	188
	66	15	16	3	159	164
	60	10	16	4	154	159
	29	6	16	5	103	112

4.2 Rank of Φ_{SNOVA}

In the linear system induced by Φ_{SNOVA}, we can use any set of 2-forms which vanish on the same oil space as the bilinear forms corresponding to the public key of SNOVA. Recall that an attacker can generate equations using $m\ell^2$ different bilinear forms that share the same oil space. These forms correspond to matrices of the type $(\mathbf{S}^a)^{\otimes n}\mathbf{P}_i(\mathbf{S}^b)^{\otimes n}$, where $0 \le a, b < \ell$ and $1 \le i \le m$. The associated 2-forms $\widehat{Q}_{i,a,b}$ associated with $(\mathbf{S}^a)^{\otimes n}\mathbf{P}_i(\mathbf{S}^b)^{\otimes n} + (\mathbf{S}^b)^{\otimes n}\mathbf{P}_i^{\intercal}(\mathbf{S}^a)^{\otimes n}$ can be used to wedge with $\widehat{\mathcal{V}}$ to generate linear equations. However, using $m\ell^2$ 2-forms introduces a scenario where many of the linear equations are dependent.

We note that at most $m\ell$ 2-forms of the form $\widehat{Q}_{i,0,b}$ corresponding to $\mathbf{P}_i(\mathbf{S}^b)^{\otimes n} + (\mathbf{S}^b)^{\otimes n}\mathbf{P}_i^{\intercal}$ can create additional linear independent equations. Indeed, observe that the equations arising from $\widehat{Q}_{i,a,b}$ can be obtained as linear combinations of the equations arising from $\widehat{Q}_{i,0,b-a}$, as follows:

$$
\begin{aligned}
0 &= \widehat{Q}_{i,0,b-a} \wedge \mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_{v\ell} \\
&= \left((\mathbf{P}_i(\mathbf{S}^{b-a})^{\otimes n} + (\mathbf{S}^{b-a})^{\otimes n}\mathbf{P}_i^{\intercal}\right) \wedge \mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_{v\ell} \\
&= \left((\mathbf{S}^a)^{\otimes n}\mathbf{P}_i(\mathbf{S}^b)^{\otimes n} + (\mathbf{S}^b)^{\otimes n}\mathbf{P}_i^{\intercal}(\mathbf{S}^a)^{\otimes n}\right) \wedge (\mathbf{S}^a)^{\otimes n}\mathbf{a}_1 \wedge \cdots \wedge (\mathbf{S}^a)^{\otimes n}\mathbf{a}_{v\ell} \\
&= \det(\mathbf{S})^{a(v\ell+2)}\widehat{Q}_{i,a,b} \wedge \mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_{v\ell}
\end{aligned}
$$

This immediately implies that all of the $m\ell^2$ 2-forms are obtainable from at most $m\ell$ 2-forms $\mathbf{P}_i(\mathbf{S}^b)^{\otimes n} + (\mathbf{S}^b)^{\otimes n}\mathbf{P}_i^{\intercal}$ where $0 \le b < \ell$. Therefore, we analyze two scenarios where the attacker selects two different sets of 2-forms to attack SNOVA via solving the kernel vector in Φ_{SNOVA}.

Scenario 1: Using $\boldsymbol{m}$ many 2-forms $\widehat{\boldsymbol{Q}}_{\boldsymbol{i},\mathbf{0},\mathbf{0}}$. In this scenario, we use m base 2-forms $\mathbf{P}_i + \mathbf{P}_i^{\intercal}$ of SNOVA and build the linear mapping

$$\begin{aligned} \Phi^{\langle m \rangle}_{\text{SNOVA}} : \bigwedge^{\ell v} \mathbb{F}_q^{\ell n} &\longrightarrow \left(\bigwedge^{\ell v+2} \mathbb{F}_q^{\ell n} \right)^m \\ \widehat{\mathcal{V}} &\longmapsto (\, \widehat{\mathcal{V}} \wedge \widehat{Q}_1, \ldots, \widehat{\mathcal{V}} \wedge \widehat{Q}_m \,) \end{aligned}$$

The dimension of $\Phi^{\langle m \rangle}_{\text{SNOVA}}$ does not change; however, the rank of $\Phi^{\langle m \rangle}_{\text{SNOVA}}$ changed. We have the following proposition:

Proposition 2. *Let* $0 \le a_i \le n - v$, $\mathbf{a} = (a_1, ..., a_\ell)$, $\sum_{i=1}^{\ell} a_i = 2r$, *and*

$$N^{\langle m \rangle}_{r,k}(t, \mathbf{z}, y; \mathbf{a}) = \left[z_1^{a_1} \cdots z_l^{a_l}\, t^{\,r} y^{\,k} \right] \left(\prod_{i=1}^{\ell} \frac{1}{1 - t\, z_i^2} \prod_{1 \le i < j \le \ell} \frac{1}{1 - t\, y\, z_i z_j} \right)^m$$

$$R^{\langle m \rangle}_{SNOVA} = \sum_{r=1}^{\lfloor \frac{\ell o}{2} \rfloor} (-1)^{\,r+1} \sum_{\substack{0 \le a_i \le o \\ \sum_i a_i = 2r}} \left(\sum_{k=0}^{r} N^{\langle 1 \rangle}_{r,k}(t, \mathbf{z}, y; \mathbf{a}) \right) \prod_{i=1}^{\ell} \binom{n}{v + a_i}$$

then

$$\operatorname{rank} \Phi^{\langle m \rangle}_{SNOVA} = \min \left\{ \binom{n}{v}^{\ell} - 1 \,,\; R^{\langle m \rangle}_{SNOVA} \right\}. \tag{9}$$

Proof. See Appendix B.

Scenario 2: Using $\boldsymbol{\ell m}$ many 2-forms $\widehat{\boldsymbol{Q}}_{\boldsymbol{i},\boldsymbol{a},\mathbf{0}}$ In this scenario, we use ℓm 2-forms $\mathbf{P}_i(\mathbf{S}^b)^{\otimes n} + (\mathbf{S}^b)^{\otimes n}\mathbf{P}_i^{\intercal}$ where $0 \le b < \ell$ of SNOVA and build the linear mapping

$$\begin{aligned} \Phi^{\langle \ell m \rangle}_{\text{SNOVA}} : \bigwedge^{\ell v} \mathbb{F}_q^{\ell n} &\longrightarrow \left(\bigwedge^{\ell v+2} \mathbb{F}_q^{\ell n} \right)^{\ell m} \\ \widehat{\mathcal{V}} &\longmapsto (\, \widehat{\mathcal{V}} \wedge \widehat{Q}_1, \ldots, \widehat{\mathcal{V}} \wedge \widehat{Q}_{\ell m} \,) \end{aligned}$$

Proposition 3. *Let* $0 \le a_i \le n - v$, $\mathbf{a} = (a_1, ..., a_\ell)$, $\sum_{i=1}^{\ell} a_i = 2r$, *and*

$$N^{\langle \ell m \rangle}_{r,k}(t, \mathbf{z}, y; \mathbf{a}) = \left[z_1^{a_1} \cdots z_l^{a_l}\, t^{\,r} y^{\,k} \right] \left(\prod_{i=1}^{\ell} \frac{1}{1 - t\, z_i^2} \prod_{1 \le i < j \le \ell} \frac{1}{(1 - t\, y\, z_i z_j)^2} \right)^m$$

$$R_{SNOVA}^{\langle \ell m \rangle} = \sum_{r=1}^{\lfloor \frac{\ell o}{2} \rfloor} (-1)^{r+1} \sum_{\substack{0 \le a_i \le o \\ \sum_i a_i = 2r}} \left(\sum_{k=0}^{r} N_{r,k}^{\langle 2 \rangle}(t, \mathbf{z}, y; \mathbf{a}) \right) \prod_{i=1}^{\ell} \binom{n}{v + a_i}$$

then

$$\operatorname{rank} \Phi_{SNOVA}^{\langle \ell m \rangle} = \min \left\{ \binom{n}{v}^{\ell} - 1 \ , \ R_{SNOVA}^{\langle \ell m \rangle} \right\}. \tag{10}$$

Proof. See Appendix C.

It should be noted that the only difference between Proposition 2 and Proposition 3 in the definition of $N_{r,k}^{\langle \ell m \rangle}(t, \mathbf{z}, y; \mathbf{a})$ is that Proposition 3 replaces the factor $(1 - t\, y\, z_i z_j)$ with $(1 - t\, y\, z_i z_j)^2$. We conducted experiments with various parameters and found that the number of linearly independent equations always matches *exactly* Formula 9 for Scenario 1 and Formula 10 for Scenario 2 (see Table 3).

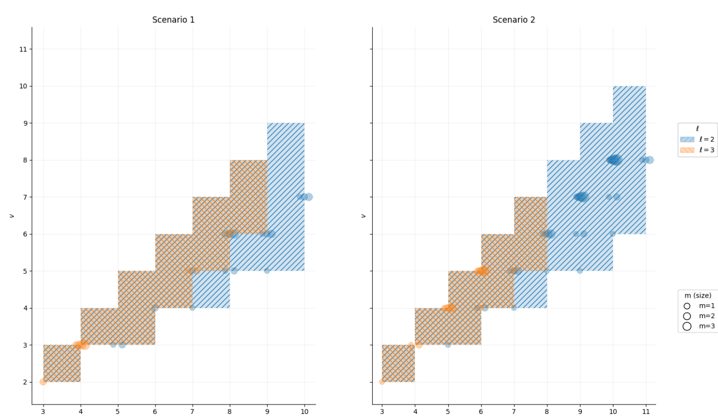

Fig. 1. The shaded area describes the (v, o, ℓ) triplets that were tested for possible m in Scenario 1 and 2. The number of linearly independent equations always matches *exactly* Formula 9 and Formula 10.

4.3 Unwanted solutions

One important factor in the wedge attack is that we want the kernel of any linear system Φ built for UOV or SNOVA to have dimension 1. However, in

Conjecture 1 when $v \geq 2m$, we can easily see that $\dim \ker \Phi_{UOV} > 1$ because

$$\left(\widehat{Q}_1 \wedge \widehat{Q}_2 \wedge \cdots \wedge \widehat{Q}_m\right) \wedge \alpha \in \ker \Phi_{UOV} \quad \text{for all } \alpha \in \bigwedge^{v-2m} \mathbb{F}^n.$$

Additionally, the oil space is not expected to be unique once $v \geq \frac{o-1}{2}m$. Such spurious solutions were identified in [27] and the author proposed restricting the attack to parameter choices satisfying

$$v < \min\left\{\frac{o-1}{2}, 2\right\} m \; .$$

Under this condition, the original wedge attack *does* not apply to SNOVA.

In our wedge attack against SNOVA, we do not expect $\left(\widehat{Q}_1 \wedge \cdots \wedge \widehat{Q}_m\right) \wedge \alpha$ to appear in the kernel. The reason is that the annihilator space $\mathcal{A}$ of $\mathcal{O}$ is highly structured with the appearance of elements from $\mathbb{F}_q[\mathbf{S}]$ while $\widehat{Q}_i \in \bigwedge^2 \mathbb{F}_q^{\ell n}$ are not. The absence of spurious solutions in our model is supported by extensive experiments conducted specifically on SNOVA parameters, where v exceeds twice the number of 2-forms (see Table 5 and Table 6). Meanwhile, the next section explores the uniqueness condition for the oil space of SNOVA, which is imposed both on v and the projection of oil coordinates.

4.4 Projection of oil coordinates

We may *project away* some oil coordinates: we keep the v vinegar coordinates together with only $o' \leq o$ oil coordinates. If we recover an o'-dimensional subspace of the *true* oil space, then the remaining $o - o'$ oil coordinates can be lifted (recovered) efficiently by linear algebra.

There is, however, a limit to how far we can project: when o' is too small, the attack can no longer reliably distinguish the genuine oil subspace from random ones, i.e., the oil space is no longer unique. In practice, the choice of o' is governed by three criteria:

1. The subspace output by the attack should be expected to lie inside the oil space, which is enforced by choosing o' large enough that the probability that such an o'-subspace exists "by chance" for a generic system is negligible.
   For $\mathbf{O} \supset \mathbf{O}' = \begin{bmatrix} \mathbf{X} \\ \mathbf{I}_{o'} \end{bmatrix} \in (\mathbb{F}_q[\mathbf{S}])^{n \times o'}$, there are $vo'\ell$ degrees of freedom. For each of the M polar forms $\mathbf{Q}_i$ (where $M = m$ or $m\ell$), $\iota(\mathbf{O}^\intercal \mathbf{Q}_i \mathbf{O}) = 0$ yields $\binom{o'\ell}{2} M$ quadratic equations over $\mathbb{F}_q$ since the $\mathbf{Q}_i$ are symmetric and have 0 on the diagonal. In order to expect $\mathbf{O}' \subset \mathbf{O}$, o' should be large enough so a solution is not expected for these equations. This is equivalent to $vo'\ell \leq \binom{o'\ell}{2} M$, so

$$v \leq \frac{o'\ell - 1}{2} M \quad \text{or} \quad o' \geq \frac{\left(\frac{2v}{M} + 1\right)}{\ell} \; .$$

2. We can use the same argument as above, but using the equations $\mathbf{O}^\intercal\mathbf{P}_i\mathbf{O}=0$, which corresponds to $m(o'\ell)^2$ equations. Thus o' must also satisfy
$$vo'\ell \leq m(o'\ell)^2 \quad \text{or} \quad o' \geq \frac{v}{m\ell}.$$
Notice that this condition is more restrictive than the first when $M = m\ell$ and $\ell \geq 3$.
3. After the projection, we must still obtain enough *independent* linear equations to recover the unknown Plücker coordinates. Equivalently, the linear system defining the candidate o'-dimensional subspace should be *overdetermined*: the number of linearly independent equations must be 1 less than the number of unknowns. Therefore, we require $R^{\langle m\rangle}_{\text{SNOVA}} \geq U_{\text{SNOVA}} - 1$ in Scenario 1 and $R^{\langle \ell m\rangle}_{\text{SNOVA}} \geq U_{\text{SNOVA}} - 1$ in Scenario 2.

Additionally, while we have parametrized our attack on the assumption that, when the oil space is represented in block form, an equal number o' of oil coordinates are projected out of each of the ℓ blocks, it is possible to consider the case that the number of coordinates projected out of each block could differ. In such case, rather than a single integer o', our attack would be parametrized by a vector $\mathbf{o}' = (o'_1, \dots, o'_l)$. However, the conditions over the uniqueness of the oil subspace need to change when using $\mathbf{o}'$. We need to replace $o'\ell$ with $\sum_{i=1}^{\ell} o'_i$. This corresponds respectively to
$$\sum_{i=1}^{\ell} o'_i \geq \frac{2v}{M} + 1 \text{ and } \sum_{i=1}^{\ell} o'_i \geq \frac{v}{m}.$$
This added flexibility could allow a projection strategy that yields a better complexity than the strategies we have thus far considered. Thus, the vector $\mathbf{o}'$ will be selected so that the attack yields the lowest cost while satisfying the conditions mentioned above.

4.5 Density of the linear system

Each equation is given by an index set $K \subset [n\ell]$ with $|K| = v\ell + 2$. In order to have a nonzero equation, at least v indices must be chosen from each block. Since there is the block-diagonal basis $\mathbf{A}_2 = \text{diag}(\mathbf{B}_1, \dots, \mathbf{B}_\ell)$ of the annihilator space $\mathcal{A}$, every nonzero $(v\ell)$-minor of $\mathbf{A}_2$ factors as a product of v-minors from the blocks; in particular, since $\mathbf{A} = \begin{bmatrix} \mathbf{I}_{v\ell} \\ -\mathbf{X}^\intercal \end{bmatrix}$, matrices $\mathbf{A}_1$ and each block of $\mathbf{A}_2$ inherit this structure, so $\mathbf{B}_i = \begin{bmatrix} \mathbf{I}_v \\ -\mathbf{X}_i^\intercal \end{bmatrix}$ for some $\mathbf{X}_i \in \mathbb{F}_{q^\ell}^{v\times o}$. As a result, each block has a v-minor with determinant 1. This yields two types for the equations:

1. $v + 2$ indices are chosen from one block, and v indices are chosen from the rest. Choosing the minors with determinant 1 from the remaining blocks

yields

$$\sum_{j<j'\in J} (\mathbf{Q}_i)_{j,j'} \det((\mathbf{B}_k)_{J\setminus\{j,j'\},*}) = 0$$

for some $k \in [\ell]$ and index set $J \subset [n]$ with $|J| = v+2$. There are $\ell\binom{n}{v+2}M$ many of these linear equation with (at most) $\binom{v+2}{2}$ nonzero terms.

2. $v+1$ indices are chosen from two distinct blocks, and v indices are chosen from the rest. Choosing the minors with determinant 1 from the remaining blocks yields

$$\sum_{j\in J \text{ and } j'\in J'} (\mathbf{Q}_i)_{j,j'} \det((\mathbf{B}_k)_{J\setminus\{j\},*}) \det((\mathbf{B}_{k'})_{J'\setminus\{j'\},*}) = 0$$

for some $k < k' \in [\ell]$ and index sets $J, J' \subset [n]$ with $|J| = |J'| = v+1$. There are $\binom{\ell}{2}\binom{n}{v+1}^2 M$ many of these bilinear equation with (at most) $(v+1)^2$ nonzero terms.

Note that, for cryptographically-relevant parameters, $(v+1)^2 > \binom{v+2}{2}$; moreover, because $2\binom{v+2}{2} > (v+1)^2$, treating all equations as having the greater density increases the complexity by less than one bit. Therefore, we use

$$\tau = (v+1)^2$$

in complexity estimation.

4.6 Complexity of the attack

We now estimate the attack complexity for concrete SNOVA parameter sets in both scenarios. Let $U_{\text{SNOVA}} = \binom{v+o}{v}^{\ell}$ denote the number of variables, and let E_{SNOVA} denote the number of linear equations in the system induced by Φ_{SNOVA}. In Scenario 1, the attacker builds the linear system using only m base 2-forms, so the expected number of *linearly independent* equations is $R^{\langle m\rangle}_{\text{SNOVA}}$ (Proposition 2). In Scenario 2, the attacker instead uses all ℓm base 2-forms, in which case the expected number becomes $R^{\langle \ell m\rangle}_{\text{SNOVA}}$ (Proposition 3). The complexity of finding a unique kernel vector of the linear system is in $\mathcal{O}(\tau E_{\text{SNOVA}} U_{\text{SNOVA}})$ operations in $\mathbb{F}_{q^\ell}$ using Wiedemann [11,18,29] algorithm, where $\tau = (v+1)^2$ is the density per row of the linear system.

After choosing the smallest possible o' per block, in accordance with Section 4.4, we observe that the difference between $\log_2(E_{\text{SNOVA}})$ and $\log_2(U_{\text{SNOVA}})$ is less than 4 bits. As in [27], we assume that the system is solvable by randomly choosing U_{SNOVA} many equations, and then the number of binary gates required for the attack against SNOVA can be computed.

$$\log_2\left(3 \cdot \left(2 \cdot (\log_2 q^\ell)^2 + \log_2 q^\ell\right) \cdot \tau \cdot U^2_{\text{SNOVA}}\right).$$

Here we assume that one multiplication in $\mathbb{F}_{q^\ell}$ costs approximately $(\log_2 q^\ell)^2$ bit operations, and we use the same order of magnitude for additions. Moreover, we assume that each field multiplication is accompanied by one field addition; this additional addition costs $\log_2 q^\ell$ bit additions. The factor 3 accounts for translating bit operations into an equivalent number of binary gates.

Table 3 summarizes our work by giving the complexity of our attack in comparison to the best attack as listed in the Round 2 specification of SNOVA [30]. The results are categorized by security levels of SNOVA under the names of SNOVA-I, SNOVA-III, and SNOVA-V.

Table 3. Bit-complexity estimates (upper bound for the base-2 logarithm of the number of binary gates required) to perform our attack against SNOVA. The column "m forms" corresponds to Scenario 1 while column "$m\ell$ forms" corresponds to Scenario 2. Note that for SNOVA concrete parameters we have $o = m$. The dash indicates that we found no o' that enables our attack. Numbers in bold refer to the best attack while italicized ones refer to the cost below the required security level.

Variants	v	m	ℓ	**Balanced projection** o'	m **forms**	o'	$m\ell$ **forms**	**Unbalanced projection** o'	m **forms**	o'	$m\ell$ **forms**	**Best attack in [30]**
SNOVA-I	37	17	2	[10,10]	148	[8,8]	*130*	[0,15]	*103*	[0,15]	***103***	153
($q = 16$)	25	8	3	[7,7,7]	149	[5,5,5]	*122*	[4,8,8]	143	[0,7,8]	***110***	166
	24	5	4	-	-	[5,5,5,5]	**155**	-	-	[5,5,5,5]	**155**	180
SNOVA-III	56	25	2	[13,13]	201	[11,11]	*181*	[0,20]	*140*	[0,20]	***140***	221
($q = 16$)	49	11	3	-	-	[11,11,11]	251	-	-	[10,11,11]	246	**220**
	37	8	4	-	-	[7,7,7,7]	223	-	-	[3,8,8,8]	**214**	257
	24	5	5	-	-	[4,4,4,4,4]	*164*	-	-	[0,5,5,5,5]	***155***	257
SNOVA-V	75	33	2	[17,17]	262	[13,13]	*222*	[0,26]	***180***	[0,26]	***180***	288
($q = 16$)	66	15	3	-	-	[14,14,14]	324	-	-	[10,15,15]	313	**293**
	60	10	4	-	-	-	-	-	-	-	-	**343**
	29	6	5	-	-	[5,5,5,5,5]	*202*	-	-	[0,5,6,6,6]	***181***	310

4.7 Modeling with quadratic equations

As mentioned in discussion about density above, we get $\ell\binom{n}{v+2}M$ equations of the form

$$\sum_{j<j'\in J} (\mathbf{Q}_i)_{j,j'} \det((\mathbf{B}_k)_{J\setminus\{j,j'\},*}) = 0,$$

which can be used to reduce the number of variables. Each block behaves like a generic UOV instance, so the number of linearly independent equations of the first type is expected to satisfy Conjecture 1. Let E_0 denote the number of these linearly independent equations for a given block. Then the number of variables is given by

$$U = \left(\binom{n}{v} - E_0\right)^\ell .$$

The rank is still expected to be $U - 1$, and the system is expected to be solvable if $U - 1$ equations are chosen at random. The drawback of this method is that eliminating variables does not preserve the sparsity of the equations. As a result, the complexity is given by $\mathcal{O}(U^\omega)$. Some complexity estimates are given in Table 4, which do not outperform the sparse complexity estimates.

5 Future work

There are several extensions of our attack, which remain outside the scope of the current state of our complexity analysis. In particular, when our attack is viewed as a quadratic system as described in Section 4.7, we have only computed concrete complexities for parameterizations of our attack that correspond to solving the system at degree-ℓ. Nonetheless, it remains a possibility that a better complexity may be obtained by solving the system at a higher degree, since doing so may allow us to use a smaller value of o' in the cases where an attack with solving degree-ℓ is possible. Allowing a higher solving degree may also allow us to solve the system in the cases where no parametrization of our attack works at degree-ℓ. Investigation of these possible refinements of our attacks will, however, be left for future work.

6 Conclusion

In this work, we revisit the wedge attack against UOV-like variants and extend this approach to a new attack on SNOVA by exploiting its block-ring structure. In addition to improving the attack complexity, the way we exploit the block-ring structure eliminates spurious solutions, which prevent the generic version of the wedge attack from working against SNOVA. Our attack breaks 6 of the 11 currently proposed SNOVA parameter sets and improves on the previous best result for an additional 2 sets; it is significantly more effective against larger ℓ values. For example, for SNOVA-V with $(v, o, \ell) = (29, 6, 5)$ the estimated security drops to 181 bits compared to 310 bits previously.

Acknowledgments The authors would like to thank the SNOVA team and Lars Ran for helpful discussions. This work was supported by the France 2030 ANR Project ANR-22-PECY-003 SecureCompute.

References

1. Aardal, M.A., Adj, G., Aranha, D.F., Basso, A., Martínez, I.A.C., Chávez-Saab, J., Santos, M.C.R., Dartois, P., Feo, L.D., Duparc, M., Eriksen, J.K., Fouotsa, T.B., , Filho, D.L.G., Hess, B., Kohel, D., Leroux, A., Longa, P., Maino, L., Meyer, M., Nakagawa, K., Onuki, H., Panny, L., Patranabis, S., Petit, C., Pope, G., Reijnders, K., Robert, D., Henríquez, F.R., Schaeffler, S., Wesolowski, B.: SQIsign: Specification Document – Round 2. Technical Report Round 2 (March 2025), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-2/spec-files/sqisign-spec-round2-web.pdf, available at NIST CSRC

2. Alagic, G., Bros, M., Ciadoux, P., Cooper, D., Dang, Q., Dang, T., Kelsey, J., Lichtinger, J., Liu, Y.K., Miller, C., Moody, D., Peralta, R., Perlner, R., Robinson, A., Silberg, H., Smith-Tone, D., Waller, N.: Status report on the first round of the additional digital signature schemes for the nist post-quantum cryptography standardization process. NIST IR **8528** (2024)
3. Benadjila, R., Bouillaguet, C., Feneuil, T., Rivain, M.: MQOM: Specification Document – Round 2. Technical Report Round 2 (March 2025), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-2/spec-files/mqom-spec-round2-web.pdf, available at NIST CSRC
4. Beullens, W.: Breaking Rainbow takes a weekend on a laptop. In: Dodis, Y., Shrimpton, T. (eds.) Advances in Cryptology - CRYPTO 2022 - 42nd Annual International Cryptology Conference, CRYPTO 2022, Santa Barbara, CA, USA, August 15-18, 2022, Proceedings, Part II. Lecture Notes in Computer Science, vol. 13508, pp. 464–479. Springer (2022). https://doi.org/10.1007/978-3-031-15979-4_16, https://doi.org/10.1007/978-3-031-15979-4_16
5. Beullens, W.: Improved cryptanalysis of SNOVA. In: Advances in Cryptology - EUROCRYPT 2025. Lecture Notes in Computer Science, vol. 15606, pp. 277–293. Springer (2025)
6. Beullens, W., Campos, F., Celi, S., Hess, B., Kannwischer, M.J.: MAYO: Specification Document – Round 2. Technical Report Round 2 (March 2025), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-2/spec-files/mayo-spec-round2-web.pdf, available at NIST CSRC
7. Beullens, W., Chen, M.S., Ding, J., Gong, B., Kannwischer, M.J., Patarin, J., Peng, B.Y., Schmidt, D., Shih, C.J., Tao, C., Yang, B.Y.: UOV: Specification document – round 2. Technical Report Round 2 (March 2025), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-2/spec-files/uov-spec-round2-web.pdf, available at NIST CSRC
8. Beullens, W., Preneel, B.: Field lifting for smaller uov public keys. In: International Conference on Cryptology in India. pp. 227–246. Springer (2017)
9. Cabarcas, D., Li, P., Verbel, J.A., Villanueva-Polanco, R.: Improved attacks for SNOVA by Exploiting Stability under a Group action. In: CRYPTO (2025)
10. Castryck, W., Decru, T.: An efficient key recovery attack on SIDH. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 423–447. Springer (2023)
11. Coppersmith, D.: Solving homogeneous linear equations over GF(2) via block Wiedemann algorithm. Mathematics of Computation **62**(205), 333–350 (1994)
12. Ding, J., Chen, M.S., Petzoldt, A., Schmidt, D., Yang, B.Y., Kannwischer, M., Patarin, J.: Rainbow: Proposal for NISTPQC: Digital signature algorithms version 3.0. Technical Report Round 3 (October 2020), https://csrc.nist.gov/Projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions, available at NIST CSRC
13. Ding, J., Deaton, J., Vishakha, Yang, B.: The nested subset differential attack - A practical direct attack against LUOV which forges a signature within 210 minutes. In: Canteaut, A., Standaert, F. (eds.) Advances in Cryptology - EUROCRYPT 2021 - 40th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Zagreb, Croatia, October 17-21, 2021, Proceedings, Part I. Lecture Notes in Computer Science, vol. 12696, pp. 329–347. Springer (2021). https://doi.org/10.1007/978-3-030-77870-5_12, https://doi.org/10.1007/978-3-030-77870-5_12

14. Furue, H., Ikematsu, Y.: A new cryptanalysis against uov-based variants mayo, QR-UOV and VOX. IEICE Trans. Fundam. Electron. Commun. Comput. Sci. **108**(3), 174–182 (2025). https://doi.org/10.1587/TRANSFUN.2024CIP0003, https://doi.org/10.1587/transfun.2024cip0003
15. Furue, H., Ikematsu, Y., Hoshino, F., Takagi, T., Kosuge, H., Yamakoshi, K., Akiyama, R., Nakamura, S., Orihara, S., Kinjo, K.: QR-UOV: Specification Document – Round 2. Technical Report Round 2 (March 2025), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/qruov-spec-round2-web.pdf, available at NIST CSRC
16. Furue, H., Ikematsu, Y., Nakamura, S., Akiyamal, R.: Improved cryptanalysis of snova by solving multi-homogeneous systems. In: Advances in Cryptology–ASIACRYPT 2025: 31st International Conference on the Theory and Application of Cryptology and Information Security, Melbourne, VIC, Australia, December 8–12, 2025, Proceedings, Part IV. p. 405. Springer Nature (2026)
17. Ikematsu, Y., Akiyama, R.: Revisiting the security analysis of SNOVA. IACR Cryptol. ePrint Arch. p. 96 (2024), https://eprint.iacr.org/2024/096
18. Kaltofen, E.: Analysis of Coppersmith's block Wiedemann algorithm for the parallel solution of sparse linear systems. Mathematics of Computation **64**(210), 777–806 (1995)
19. Kipnis, A., Patarin, J., Goubin, L.: Unbalanced Oil and Vinegar Signature Schemes. In: Stern, J. (ed.) Advances in Cryptology - EUROCRYPT '99, International Conference on the Theory and Application of Cryptographic Techniques, Prague, Czech Republic, May 2-6, 1999, Proceeding. Lecture Notes in Computer Science, vol. 1592, pp. 206–222. Springer (1999), https://doi.org/10.1007/3-540-48910-X_15
20. Li, P., Ding, J.: Cryptanalysis of the SNOVA signature scheme. In: Saarinen, M.O., Smith-Tone, D. (eds.) Post-Quantum Cryptography - 15th International Workshop, PQCrypto 2024, Oxford, UK, June 12-14, 2024, Proceedings, Part II. Lecture Notes in Computer Science, vol. 14772, pp. 79–91. Springer (2024), https://doi.org/10.1007/978-3-031-62746-0_4
21. Maino, L., Martindale, C.: An attack on SIDH with arbitrary starting curve. IACR Cryptol. ePrint Arch. p. 1026 (2022), https://eprint.iacr.org/2022/1026
22. Nakamura, S., Tani, Y., Furue, H.: Lifting approach against the SNOVA scheme. IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences p. 2024EAP1124 (2025)
23. NIST: Call for Additional Digital Signature Schemes for the Post-quantum Cryptography Standardization Process. NIST CSRC (2022), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/call-for-proposals-dig-sig-sept-2022.pdf
24. Patarin, J., Cogliati, B., Faugère, J.C., Fouque, P.A., Goubin, L., Larrieu, R., Macario-Rat, G., Minaud, B.: VOX specification v1.0. Round 1 Additional Signatures, NIST Post-Quantum Cryptography: Digital Signature Schemes (Jun 2023), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-1/spec-files/vox-spec-web.pdf, version 1.0
25. Ran, L.: Round 2 (additional signatures) official comment: UOV, MAYO, SNOVA. https://groups.google.com/a/list.nist.gov/g/pqc-forum/c/NNv4C5PRIjs (May 2025), posted to the pqc-forum@list.nist.gov mailing list
26. Ran, L.: Wedges, oil, and vinegar – an analysis of UOV in characteristic 2. Cryptology ePrint Archive, Paper 2025/1143 (2025), https://eprint.iacr.org/2025/1143

27. Ran, L.: Wedges, Oil, and Vinegar: A New Algorithm for UOV in Characteristic 2. Rump Session, Eurocrypt 2025 (2025), https://larsmath.github.io/assets/pdf/rumpsessioneurocrypt25.pdf, presented May 6, 2025
28. Robert, D.: Breaking SIDH in polynomial time. In: Annual International Conference on the Theory and Applications of Cryptographic Techniques. pp. 472–503. Springer (2023)
29. Villard, G.: A study of Coppersmith's block Wiedemann algorithm using matrix polynomials. Citeseer (1997)
30. Wang, L.C., Chou, C.Y., Ding, J., Kuan, Y.L., Leegwater, J.A., Li, M.S., Tseng, B.S., Tseng, P.E., Wang, C.C.: SNOVA: Specification Document – Round 2. Technical Report Round 2 (March 2025), https://csrc.nist.gov/csrc/media/Projects/pqc-dig-sig/documents/round-2/spec-files/snova-spec-round2-web.pdf, available at NIST CSRC
31. Yokonuma, T.: Tensor spaces and exterior algebra. No. 108, American Mathematical Soc. (1992)

A Arguments for Conjecture 1

Let $K = \mathbb{F}_{2^k}$ and $F = K^n = \mathcal{A} \oplus \mathcal{B}$ where $\mathcal{A}$ is the Annihilator space previously defined and $\mathcal{B}$ is a subspace that completes the full basis where $\dim \mathcal{A} = v$, $\dim \mathcal{B} = n - v$, and $n < 2v$. For $a \leq v$, let

$$N_a := \bigwedge^a \mathcal{A} \wedge \bigwedge^{v+a} F := \operatorname{span}_K\{\alpha \wedge \beta : \alpha \in \bigwedge^a \mathcal{A}, \beta \in \bigwedge^{v+a} F\} \subseteq \bigwedge^{v+2a} F.$$

Fix m 2-forms $\widehat{Q}_1, ..., \widehat{Q}_m \in \mathcal{A} \wedge F$ where $\mathcal{A} \wedge F$ denotes the subspace of $F \wedge F$ spanned by the vectors $v \wedge f$, for $v \in \mathcal{A}$ and $f \in F$. Let $E = \mathbb{F}_q^m$ be a vector space with canonical basis $x_1, \ldots, x_m$. Intuitively, we think of E as "indexing" the previous forms, via the map $E \to \mathcal{A} \wedge F$ sending $(\lambda_i)_{i=1}^m$ to $\sum_{i=1}^m \lambda_i \widehat{Q}_i$. Let $\operatorname{Sym}^a E$ denote the elements of degree a in the (graded) symmetric algebra, equipped with the product "$\cdot$". This is isomorphic to the homogeneous polynomials of degree a over $\dim(E)$ indeterminates; we use tensor terminology only because it mixes more naturally with the rest of the notation. We define a vector space

$$C^a = \operatorname{Sym}^a E \otimes N_a$$

and the linear map

$$\begin{aligned} \delta_a : C^a &\longrightarrow C^{a+1} \\ \eta \otimes \mu &\longmapsto \sum_{i=1}^m (x_i \cdot \eta) \otimes (\widehat{Q}_i \wedge \mu). \end{aligned}$$

We introduce the following lemma:

Lemma 3.

$$\bigwedge^i \mathcal{A} \wedge \bigwedge^{v+i} F = \bigwedge^{v+2i} F$$

Proof.

$$\begin{aligned}\bigwedge^{i}\mathcal{A}\wedge\bigwedge^{v+i}F &= \bigwedge^{i}\mathcal{A}\wedge\bigwedge^{v+i}(\mathcal{A}\oplus\mathcal{B}) = \bigwedge^{i}\mathcal{A}\wedge\bigoplus_{k_1+k_2=v+i}\bigwedge^{k_1}\mathcal{A}\otimes\bigwedge^{k_2}\mathcal{B}\\ &= \bigoplus_{k_1+k_2=v+i}\bigwedge^{i+k_1}\mathcal{A}\otimes\bigwedge^{k_2}\mathcal{B} = \bigoplus_{\substack{k_1+k_2=v+2i\\ k_1\geq i}}\bigwedge^{k_1}\mathcal{A}\otimes\bigwedge^{k_2}\mathcal{B}\\ &= \bigoplus_{\substack{k_1+k_2=v+2i\\ k_2\leq v+i}}\bigwedge^{k_1}\mathcal{A}\otimes\bigwedge^{k_2}\mathcal{B}\\ &= \bigwedge^{v+2i}F \text{ because } \dim\mathcal{B}=n-v<v+i\ \ \forall i\geq 0 \text{ in our setting.}\end{aligned}$$

$\square$

From Lemma 3, we can see that

$$C^a = \mathrm{Sym}^a E\otimes\bigwedge^{v+2a}F \quad\text{and}\quad \dim C^a = \binom{m+a-1}{a}\binom{v+o}{v+2a}.$$

Let us define the linear map

$$\begin{aligned}\delta_a : C^a &\longrightarrow C^{a+1}\\ \eta\otimes\mu &\longmapsto \sum_{i=1}^{m}(x_i\cdot\eta)\otimes(\widehat{Q}_i\wedge\mu).\end{aligned}$$

The goal is to build a cochain complex of $K^\bullet(C^\bullet,\widehat{Q}_\bullet)$. In order to do that, we need the following lemma:

Lemma 4. *For any 2-form $\widehat{Q}$ over a finite field of characteristic 2, we have*

$$\widehat{Q}\wedge\widehat{Q}=0.$$

Proof. Let $\widehat{Q}=\sum_{1\leq i<j\leq n}\mathbf{Q}_{i,j}\,\mathbf{e}_i\wedge\mathbf{e}_j\in\bigwedge^2\mathbb{F}_q^n$ where $\mathbf{e}_i$ is the canonical basis for $\mathbb{F}_q^n$. Then

$$\widehat{Q}\wedge\widehat{Q}=\sum_{\substack{1\leq i<j\leq n\\ 1\leq k<\ell\leq n}}(\mathbf{Q}_{i,j}\,\mathbf{e}_i\wedge\mathbf{e}_j)\wedge(\mathbf{Q}_{k,\ell}\,\mathbf{e}_k\wedge\mathbf{e}_\ell)$$

When $\{i,j\}\neq\{k,\ell\}$, these terms come in pairs. Since $\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\wedge\mathbf{e}_\ell=\mathbf{e}_k\wedge\mathbf{e}_\ell\wedge\mathbf{e}_i\wedge\mathbf{e}_j$, the coefficient of $\mathbf{e}_i\wedge\mathbf{e}_j\wedge\mathbf{e}_k\wedge\mathbf{e}_\ell$ is 0 over characteristic 2. When $\{i,j\}=\{k,\ell\}$, these terms are 0 since $\mathbf{e}_i\wedge\mathbf{e}_i=0$. $\square$

Using Lemma 4, we can see that, over a finite field of characteristic 2

$$\delta_a \circ \delta_{a-1} = \sum_{i \neq j} (\eta_i \cdot \eta_j) \otimes (\widehat{Q}_i \wedge \widehat{Q}_j \wedge \mu) = \sum_{i<j} (2\eta_i \cdot \eta_j) \otimes (\widehat{Q}_i \wedge \widehat{Q}_j \wedge \mu) = 0 \quad \forall a \; .$$

Thus, we have the necessary condition to build a cochain complex below

$$0 \longrightarrow C^0 \xrightarrow{\delta_0} C^1 \longrightarrow \cdots \xrightarrow{\delta_{m-1}} C^m \longrightarrow ... \longrightarrow 0 \; .$$

We introduce the following assumption whose the validity is supported by our experimental evidence.

Assumption 1 *Let* $r = \lfloor \frac{o}{2} \rfloor$. *The cochain complex*

$$0 \longrightarrow C^0 \xrightarrow{\delta_0} C^1 \xrightarrow{\delta_1} \cdots \xrightarrow{\delta_{r-1}} C^r \longrightarrow 0$$

is exact in every positive degree. Equivalently, the only possible nonzero cohomology is in degree 0.

Under this assumption, we obtain

$$\begin{aligned}
& \dim \operatorname{Im} \delta_{a-1} = \dim \ker \delta_a \quad \forall a \geq 1 \\
\Longrightarrow \; & \dim \operatorname{Im} \delta_{a-1} = \dim C^a - \dim \operatorname{Im} \delta_a \quad \forall a \geq 1 \\
\Longrightarrow \; & \operatorname{rank} \delta_0 = \sum_{i=1}^{k} (-1)^{i+1} \dim C^i \quad \forall 1 \leq k < \lfloor \frac{o}{2} \rfloor < \frac{v}{2} \\
\Longrightarrow \; & \operatorname{rank} \delta_0 = \sum_{i=1}^{\lfloor \frac{o}{2} \rfloor} (-1)^{i+1} \binom{m+i-1}{i} \binom{v+o}{v+2i} \; .
\end{aligned}$$

B Proof of Proposition 2

Again, let $F = (\mathbb{F}_{2^k})^{\ell n} = \mathcal{A} \oplus \mathcal{B}$ where $\dim \mathcal{A} = lv$, $\dim \mathcal{B} = \ell(n-v)$. Recall that from Proposition 1 we know that $F \cong \bigoplus_{i=1}^{\ell} F_i = \bigoplus_{i=1}^{\ell} (\mathcal{A}_i \oplus \mathcal{B}_i)$ and $\mathcal{A} \cong \bigoplus_{i=1}^{\ell} \mathcal{A}_i$ where $F_i \cong \mathbb{F}_{2^k}^n, \mathcal{A}_i \cong \mathbb{F}_{2^k}^v$, then

$$\begin{aligned}
\bigwedge^g \mathcal{A} \wedge \bigwedge^h F &= \bigwedge^g \left(\bigoplus_{i=1}^{\ell} \mathcal{A}_i \right) \wedge \bigwedge^h \left(\bigoplus_{i=1}^{\ell} F_i \right) \\
&= \left(\bigoplus_{\Sigma g_i = g} \bigotimes_{i=1}^{\ell} \bigwedge^{g_i} \mathcal{A}_i \right) \wedge \left(\bigoplus_{\Sigma h_i = h} \bigotimes_{i=1}^{\ell} \bigwedge^{h_i} F_i \right)
\end{aligned}$$

We have the following lemma:

Lemma 5. *Let* $G = \bigotimes_{i=1}^{\ell} \bigwedge^{v} F_i$*, i.e., the "balanced" subspace of* $\bigwedge^{\ell v} F$*, then we have*

$$\dim\Big(\bigwedge^{r}\mathcal{A} \wedge \bigwedge^{r} F \wedge G\Big) = \sum_{\substack{0\le a_i\le n-v \\ \sum_i a_i = 2r}} \prod_{i=1}^{\ell} \binom{n}{v+a_i}. \tag{11}$$

Proof.

$$\begin{aligned}
\bigwedge^{r}\mathcal{A} \wedge \bigwedge^{r} F \wedge G &= \left(\bigoplus_{\substack{0\le a_i\le v \\ \sum_i a_i = r}} \bigotimes_{i=1}^{\ell} \bigwedge^{a_i}\mathcal{A}_i\right) \wedge \left(\bigoplus_{\substack{0\le b_i\le n \\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \bigwedge^{v+b_i} F_i\right) \\
&= \bigoplus_{\substack{0\le a_i\le v\ ,\ 0\le b_i\le n \\ \sum_i a_i = r\ ,\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \left(\bigwedge^{a_i}\mathcal{A}_i \wedge \bigwedge^{v+b_i} F_i\right) \\
&= \bigoplus_{\substack{0\le a_i\le v\ ,\ 0\le b_i\le n \\ \sum_i a_i = r\ ,\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \left(\bigwedge^{a_i}\mathcal{A}_i \wedge \bigwedge^{v+b_i} (\mathcal{A}_i \oplus \mathcal{B}_i)\right) \\
&= \bigoplus_{\substack{0\le a_i\le v\ ,\ 0\le b_i\le n \\ \sum_i a_i = r\ ,\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \left(\bigwedge^{a_i}\mathcal{A}_i \wedge \bigoplus_{k_{i1}+k_{i2}=v+b_i} \bigwedge^{k_{i1}}\mathcal{A}_i \otimes \bigwedge^{k_{i2}}\mathcal{B}_i\right) \\
&= \bigoplus_{\substack{0\le a_i\le v\ ,\ 0\le b_i\le n \\ \sum_i a_i = r\ ,\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \left(\bigoplus_{k_{i1}+k_{i2}=v+b_i} \left(\bigwedge^{a_i+k_{i1}}\mathcal{A}_i \otimes \bigwedge^{k_{i2}}\mathcal{B}_i\right)\right) \\
&= \bigoplus_{\substack{0\le a_i\le v\ ,\ 0\le b_i\le n \\ \sum_i a_i = r\ ,\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \left(\bigoplus_{\substack{k_{i1}+k_{i2}=a_i+b_i+v \\ k_{i1}\ge a_i}} \left(\bigwedge^{k_{i1}}\mathcal{A}_i \otimes \bigwedge^{k_{i2}}\mathcal{B}_i\right)\right) \\
&= \bigoplus_{\substack{0\le a_i\le v\ ,\ 0\le b_i\le n \\ \sum_i a_i = r\ ,\ \sum_i b_i = r}} \bigotimes_{i=1}^{\ell} \bigwedge^{a_i+b_i+v} F_i \\
&= \bigoplus_{\substack{0\le a_i\le n-v \\ \sum_i a_i = 2r}} \bigotimes_{i=1}^{\ell} \bigwedge^{a_i+v} F_i\ ,
\end{aligned}$$

then

$$\dim\Big(\bigwedge^r \mathcal{A} \wedge \bigwedge^r F \wedge G\Big) = \sum_{\substack{0\le a_i \le n-v \\ \sum_i a_i = 2r}} \prod_{i=1}^{\ell} \binom{n}{v+a_i}.$$

□

We fix the m 2-forms $\widehat{Q}_1, ..., \widehat{Q}_m \in \mathcal{A} \wedge F \subset \bigwedge^2 F$ spanned by the vectors $v \wedge f$, for $v \in \mathcal{A}$ and $f \in F$. Here, we call $\binom{n}{v+a_i}$ a block and o is the capacity of such block that means $a_i \le o$. Because G is now split into l blocks where wedging by any fixed $\widehat{Q}_i$ either adds $a_i = +2$ in a single block (a "double") or adds $a_i, a_j = +1$ in two distinct blocks (a "split"), indexing $\widehat{Q}_i$ with $E = \mathbb{F}_q^m$ is not sufficient; thus, we have to index them with $E = \mathrm{Sym}^2(\mathbb{F}_q^\ell)^m$. Furthermore, among ℓ single blocks, no block can exceed $a_i \le n-v$. That means to encode the behaviour of the wedges, we need the variables z_i to record block increments, the variable m to record which $\widehat{Q}_i$ is being used, and the marker y to record the splits. Therefore, for a single $\widehat{Q}$, the ordinary generating series is

$$1 + tz_i^2 + t^2 z_i^4 + \cdots = \frac{1}{1 - tz_i^2}.$$

$$1 + tyz_iz_j + (tyz_iz_j)^2 + \cdots = \frac{1}{1 - tyz_iz_j}.$$

$$G(t, \mathbf{z} = (z_1, ..., z_l), y) = \prod_{i=1}^{\ell} \frac{1}{1 - tz_i^2} \cdot \prod_{1 \le i < j \le \ell} \frac{1}{1 - tyz_iz_j}.$$

Over all m forms these choices multiply, hence the number of degree-r selections that (i) use exactly k splits and (ii) land in occupancy $\mathbf{a} = (a_1, ..., a_l)$ where $\sum_{i=1}^{\ell} a_i = 2r$ with $0 \le a_i \le n - v$ equals

$$N_{r,k}(t, \mathbf{z}, y; \mathbf{a}) = \left[\mathbf{z}^{\mathbf{a}}\, t^r y^k\right]\ G(t, \mathbf{z}, y)^m \tag{12}$$

Now, we define a bigrading vector space

$$C^a = \mathrm{Sym}^{2m}\, \mathbb{F}_q^\ell \otimes \left(\bigwedge^a \mathcal{A} \wedge \bigwedge^a F \wedge G\right)$$

and the differential

$$\delta(\eta \otimes \mu) \coloneqq \sum_{i=1}^{m} (x_i \cdot \eta) \otimes (\widehat{Q}_i \wedge \mu).$$

Note that we still have we have $\delta^2 \equiv 0$. Therefore, we fix l, v and then build a cochain complex

$$0 \longrightarrow C^0 \xrightarrow{\delta_0} C^1 \xrightarrow{\delta_1} \cdots C^m \xrightarrow{\delta_m} ... \longrightarrow 0 \,.$$

Here, every step raises the $\left(\bigwedge^{\bullet} \mathcal{A} \wedge \bigwedge^{\bullet} F \wedge G\right)$-degree by $+2$.

Projecting to the $\mathbf{a}$–summand of $\bigwedge^{r} \mathcal{A} \wedge \bigwedge^{v\ell+r} F$, the span contributed from E^r has dimension at most

$$\sum_{k=0}^{r} N_{r,k}(t, \mathbf{z}, y; \mathbf{a})$$

while the exterior factor has dimension given by Equation (11). Under Assumption 1 with $r = \lfloor \frac{\ell o}{2} \rfloor$, we have

$$\operatorname{rank} \varPhi_{\text{SNOVA}} = \sum_{r=1}^{\lfloor \frac{\ell o}{2} \rfloor} (-1)^{r+1} \sum_{\substack{\mathbf{a} \in \{0,\dots,o\}^\ell \\ \sum_i a_i = 2r}} \left(\sum_{k=0}^{r} N_{r,k}(t, \mathbf{z}, y; \mathbf{a}) \right) \prod_{i=1}^{\ell} \binom{v+o}{v+a_i}.$$

A toy example We take $n = 8$, $v = 5$, $m = 2$, $\ell = 2$, $o = n - v = 3$. Thus there are two blocks, each with *capacity* at most $o = 3$. Let z_1, z_2 mark the occupancies of blocks $1, 2$, let t mark how many 2-forms are wedged, and let $\mathbf{v}$ mark the number of *splits* (cross-block pairs).

1. **Generating series.**
   Within a single 2-form, a contribution can be:
   - *within block i*: contributes z_i^2 and uses one wedge, hence weight tz_i^2;
   - *across blocks* 1 *and* 2: contributes $z_1 z_2$, uses one wedge, and creates one split, hence weight $t\mathbf{v}z_1 z_2$.

   Therefore the geometric series for these choices are

   $$\begin{aligned} 1 + tz_i^2 + t^2 z_i^4 + \cdots &= \frac{1}{1 - tz_i^2} \\ 1 + t\mathbf{v}z_1 z_2 + (t\mathbf{v}z_1 z_2)^2 + \cdots &= \frac{1}{1 - t\mathbf{v}z_1 z_2}. \end{aligned}$$

   For one 2-form,

   $$F_1(z_1, z_2; t, \mathbf{v}) = \frac{1}{(1 - tz_1^2)(1 - tz_2^2)(1 - t\mathbf{v}z_1 z_2)}. \tag{13}$$

   For $m = 2$ independent 2-forms,

   $$F(z_1, z_2; t, \mathbf{v}) \;=\; F_1(z_1, z_2; t, \mathbf{v})^2. \tag{14}$$

Capacity ≤ 3 means: after expanding in z_1, z_2, keep only monomials $z_1^x z_2^y$ with $0 \leq x, y \leq 3$, i.e. apply the truncation operator

$$[z_1^{\leq 3} z_2^{\leq 3}](\cdot).$$

2. **Coefficient extraction for $r = 1$: $[t^1 \mathbf{v}^k]$.**
From (13),
$$[t]\, F_1 \ = \ z_1^2 + z_2^2 + \mathbf{v} z_1 z_2.$$
Using (14), for $m = 2$,
$$[t]\, F \ = \ 2(z_1^2 + z_2^2 + \mathbf{v} z_1 z_2).$$
All these monomials respect the capacity bound (≤ 3), so truncation does nothing. Grouping by powers of $\mathbf{v}$ (number of splits) gives:
$$\mathbf{v}^0 : \ 2z_1^2 + 2z_2^2, \qquad \mathbf{v}^1 : \ 2z_1 z_2.$$
Hence the *counts* (sum of coefficients after truncation) are
$$k = 0 : \ 4, \qquad k = 1 : \ 2.$$

3. **Coefficient extraction for $r = 2$: $[t^2 \mathbf{v}^k]$.**
For one 2-form,
$$[t^2]\, F_1 = z_1^4 + z_2^4 + z_1^2 z_2^2 + \mathbf{v}(z_1^3 z_2 + z_1 z_2^3) + \mathbf{v}^2 z_1^2 z_2^2.$$
For two 2-forms,
$$[t^2]\, F = 2[t^2]\, F_1 + \big([t]\, F_1\big)^2 = 3z_1^4 + 4z_1^2 z_2^2 + 3z_2^4 + \mathbf{v}\big(4z_1^3 z_2 + 4z_1 z_2^3\big) + \mathbf{v}^2 3z_1^2 z_2^2.$$
Apply capacity ≤ 3: drop z_1^4 and z_2^4. The truncated part is
$$[z_1^{\leq 3} z_2^{\leq 3}]\, [t^2]\, F = \mathbf{v}^0(4z_1^2 z_2^2) + \mathbf{v}^1(4z_1^3 z_2 + 4z_1 z_2^3) + \mathbf{v}^2(3z_1^2 z_2^2).$$
So the counts are
$$k = 0 : \ 4, \qquad k = 1 : \ 8, \qquad k = 2 : \ 3.$$

4. **Coefficient extraction for $r = 3$: $[t^3 \mathbf{v}^k]$.**
Under capacity ≤ 3, the only monomial with *both* exponents ≤ 3 at total wedge degree 3 is $z_1^3 z_2^3$. Inside $[t^3]F_1$, the capacity-feasible contributions to $z_1^3 z_2^3$ are
$$[t^3]\, F_1 \ \supset \ \big(\mathbf{v} + \mathbf{v}^3\big)\, z_1^3 z_2^3 \quad (+ \text{ terms violating the cap}).$$
Using $F = F_1^2$,
$$[t^3]\, F = 2[t^3]\, F_1 \ + \ 2\big([t^2]\, F_1\big)\big([t]\, F_1\big).$$

Keeping only the $z_1^3 z_2^3$ terms (all others are truncated), one obtains

$$[z_1^{\leq 3} z_2^{\leq 3}]\,[t^3]\,F \;\supset\; (8\mathbf{v} + 4\mathbf{v}^3)\, z_1^3 z_2^3.$$

Hence the counts are

$$k = 1:\; 8, \qquad k = 3:\; 4.$$

5. **Assembling the rank via the generating series.**
To sum over all split counts k, set $\mathbf{v} = 1$. Each occupancy (x, y) is then weighted by

$$\binom{v+o}{v+x}\binom{v+o}{v+y}, \qquad \text{with alternating sign in } r.$$

Here $v = 5$, $o = 3$, so $v + o = 8$. Define the weight polynomial

$$B(z) = \sum_{x=0}^{3} \binom{8}{5+x} z^x = 56 + 28z + 8z^2 + z^3.$$

Write $\langle \cdot, \cdot \rangle$ for the coefficientwise inner product in z_1, z_2:

$$\Big\langle \sum_{x,y} a_{x,y} z_1^x z_2^y,\ \sum_{x,y} b_{x,y} z_1^x z_2^y \Big\rangle \;=\; \sum_{x,y} a_{x,y} b_{x,y}.$$

Then the rank is computed as

$$\text{rank} = \sum_{r=1}^{3} (-1)^{r+1}\,[t^r] \Big\langle [z_1^{\leq 3} z_2^{\leq 3}]\, F(z_1, z_2; t, \mathbf{v}{=}1),\ B(z_1)B(z_2) \Big\rangle.$$

Evaluating the three r-layers above yields the subtotals

$$r = 1:\; +3360, \qquad r = 2:\; -672, \qquad r = 3:\; +12,$$

and therefore

$$\boxed{\text{rank} = 3360 - 672 + 12 = 2700}$$

which matches the observed rank for $(n, v, m, \ell) = (8, 5, 2, 2)$ in Table 5.

C Proof of proposition 3

Because $\mathbf{S}$ is diagonalizable, we have

$$\mathbf{S} = \mathbf{UDU}^{-1} \quad \text{where } \mathbf{D} = \mathrm{Diag}\,(\lambda_1, ..., \lambda_l)$$

Let

$$\mathbf{\Pi}_r = \mathbf{U}\mathbf{E}_r\mathbf{U}^{-1} \quad \text{where } \mathbf{E} = \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} \quad \text{with 1 in position } r$$

then we have $\mathbf{\Pi}_r^2 = \mathbf{\Pi}_r$, $\mathbf{\Pi}_r\mathbf{\Pi}_s = 0$ and $\sum_{i=1}^{\ell} \lambda_i \mathbf{\Pi}_i = \mathbf{S}$. By consequence, we have

$$\mathbf{S}^b = \sum_{i=1}^{\ell} \lambda_i^b \mathbf{\Pi}_i \ .$$

Let $Q^{\langle b \rangle} = \widehat{Q}_{\bullet,b,0}$ be a 2-form scaled by the matrix $(\mathbf{I}_n \otimes \mathbf{S}^b)$, then we have

$$\begin{aligned} Q^{\langle b \rangle} &= (\mathbf{I}_n \otimes S^b)\mathbf{P} + \mathbf{P}^\intercal(\mathbf{I}_n \otimes S^b) \\ &= (\mathbf{I}_n \otimes (\sum_{i=1}^{\ell} \lambda_i^b \mathbf{\Pi}_i))\mathbf{P} + \mathbf{P}^\intercal(\mathbf{I}_n \otimes (\sum_{i=1}^{\ell} \lambda_i^b \mathbf{\Pi}_i)) \\ &= \sum_{i=1}^{\ell} \bigwedge_i^b ((\mathbf{I}_n \otimes \mathbf{\Pi}_i)\mathbf{P} + \mathbf{P}^\intercal(\mathbf{I}_n \otimes \mathbf{\Pi}_i)) \\ &= \sum_{i=1}^{\ell} \bigwedge_i^b \mathbf{Z}^{\langle i \rangle} \quad \text{where } \mathbf{Z}^{\langle i \rangle} = ((\mathbf{I}_n \otimes \mathbf{\Pi}_i)\mathbf{P} + \mathbf{P}^\intercal(\mathbf{I}_n \otimes \mathbf{\Pi}_i)) \ . \end{aligned}$$

Let $\mathbf{V}$ be the Vandermonde $\mathbf{V}_{b,i} = \lambda_i^b$, then

$$\begin{pmatrix} Q^{\langle 1 \rangle} \\ Q^{\langle 2 \rangle} \\ \vdots \\ Q^{\langle \ell \rangle} \end{pmatrix} = \begin{pmatrix} \lambda_1 & \cdots & \lambda_\ell \\ \lambda_1^2 & \cdots & \lambda_\ell^2 \\ \vdots & \ddots & \vdots \\ \lambda_1^\ell & \cdots & \lambda_\ell^\ell \end{pmatrix} \begin{pmatrix} Z^{\langle 1 \rangle} \\ Z^{\langle 2 \rangle} \\ \vdots \\ Z^{\langle \ell \rangle} \end{pmatrix} = V \begin{pmatrix} Z^{\langle 1 \rangle} \\ Z^{\langle 2 \rangle} \\ \vdots \\ Z^{\langle \ell \rangle} \end{pmatrix} .$$

By consequence, we have

$$\begin{pmatrix} Q_i^{\langle 1 \rangle} \\ Q_i^{\langle 2 \rangle} \\ \vdots \\ Q_i^{\langle \ell \rangle} \end{pmatrix} = V \begin{pmatrix} Z_i^{\langle 1 \rangle} \\ Z_i^{\langle 2 \rangle} \\ \vdots \\ Z_i^{\langle \ell \rangle} \end{pmatrix} \qquad \forall i \in \{1, ..., m\} \ .$$

Since the λ_i are distinct in the splitting field, V is invertible, so the ℓ-tuple $(Q^{\langle b \rangle})_i$ spans the same ℓ-dimensional space as $(Z^{\langle b \rangle})_i$.

For a fixed ℓv-form $\widehat{\mathcal{V}}$ in $\bigwedge^{\ell v} F$, let φ be the linear mapping

$$\begin{aligned} \varphi : \left(\bigwedge^{2} F\right)^{m} &\longrightarrow \left(\bigwedge^{\ell v+2} F\right)^{m} \\ (\widehat{Q}_1, \ldots, \widehat{Q}_m) &\longmapsto (\widehat{\mathcal{V}} \wedge \widehat{Q}_1, \ldots, \widehat{\mathcal{V}} \wedge \widehat{Q}_m) \end{aligned}$$

then we have

$$\varphi \begin{pmatrix} Q_i^{\langle 1\rangle} \\ Q_i^{\langle 2\rangle} \\ \vdots \\ Q_i^{\langle \ell\rangle} \end{pmatrix} = \varphi \begin{pmatrix} Z_i^{\langle 1\rangle} \\ Z_i^{\langle 2\rangle} \\ \vdots \\ Z_i^{\langle \ell\rangle} \end{pmatrix} .$$

In Proposition 1, we proved that, up to some change-of-basis, the vinegar subspace achieves the block diagonal form

$$\mathrm{diag}(\mathbf{B}_1, \mathbf{B}_2, ..., \mathbf{B}_\ell).$$

Again, we consider the linear map corresponding to SNOVA where each F_i corresponds to $\begin{pmatrix}0 \cdots \mathbf{B_i} \cdots 0\end{pmatrix}^{\intercal}$, then we build the same cochain complex as in Appendix B with following differential

$$\begin{aligned} \delta : \bigotimes_{j=1}^{\ell} \bigwedge^{v} F_j &\longrightarrow \bigoplus_{\sum a_j = 2} \bigotimes_{j=1}^{\ell} \bigwedge^{v+a_j} F_j \\ \mu &\longmapsto \mu \wedge \widehat{Q} \end{aligned}$$

Because $\mathbf{\Pi}_i$ is the projector onto F_i, hence $\mathbf{\Pi}_i|_{F_i} = \mathrm{Id}$ and $\mathbf{\Pi}_i|_{F_j} = 0 \ \ (j \neq i)$. Let B_P be a bilinear form

$$B_P : F \times F \to k, \quad B_P(x, y) = x^{\intercal} P y.$$

We need two following lemmas:

Lemma 6.

$$Z^{\langle i\rangle}|_{\bigwedge^2 F_j} = 0 \textit{ for } j \neq i \ .$$

Proof. The bilinear form corresponding to $Z^{\langle i\rangle}$ is

$$Z^{\langle i\rangle}(x, y) = B_P(\mathbf{\Pi}_i x, y) + B_P(x, \mathbf{\Pi}_i y) \ .$$

Then, $\forall x, y \in F_j$ where $j \neq i$, we have $\mathbf{\Pi}_i x = \mathbf{\Pi}_i y = 0 \implies Z^{\langle i\rangle}(x, y) = 0$. So the restriction of $Z^{\langle i\rangle}$ to $F_j \times F_j$ is identically zero when $j \neq i$. Equivalently, $Z^{\langle i\rangle}|_{\bigwedge^2 F_j} = 0 \quad \text{for } j \neq i$. □

Lemma 7.

$$Z^{\langle i\rangle}|_{F_i \wedge F_j} = B_P|_{F_i \times F_j} \textit{ for all } j \neq i \ .$$

Proof. Indeed, restrict to $x \in F_i, y \in F_j$. Using $\mathbf{\Pi}_i x = x$, $\mathbf{\Pi}_i y = 0$ and $\mathbf{\Pi}_j x = 0$, $\mathbf{\Pi}_j y = y$, we get

$$Z^{\langle i \rangle}(x,y) = B_P(\mathbf{\Pi}_i x, y) + B_P(x, \mathbf{\Pi}_i y) = B_P(x,y),$$
$$Z^{\langle j \rangle}(x,y) = B_P(\mathbf{\Pi}_j x, y) + B_P(x, \mathbf{\Pi}_j y) = B_P(x,y).$$

So both projectors i and j hit the same cross-block wedge space $F_i \wedge F_j$. □

Recall that with the block splitting $F = \bigoplus_{j=1}^{\ell} F_j$, every 2-form are still decomposed into: Type 1 ("double") $\bigwedge^2 F_i$, and Type 2 ("split") $F_i \wedge F_j$ with $i < j$ (see Section 4.5). According to Lemma 6, for $j \neq i$, $(Z^{\langle i \rangle})|_{F_j \wedge F_j} = 0$, so the $\bigwedge^2 F_j$ piece only exists when $j = i$. Thus, for each i, each $Z^{\langle i \rangle}$ provides exactly one "double channel" $\bigwedge^2 F_i$. However, per Lemma 7, for each fixed pair $i < j$, *for a single base form* k, we have two independent ways to realize a split into (i,j): one via $Z_k^{\langle i \rangle}$ and one via $Z_k^{\langle j \rangle}$. Therefore, similar to Appendix B, the "double" part contributes a factor

$$\frac{1}{1 - tz_i^2}$$

while the "split" part contributes a factor

$$\left(\frac{1}{1 - tyz_iz_j}\right) \times \left(\frac{1}{1 - tyz_iz_j}\right) = \frac{1}{(1 - tyz_iz_j)^2}.$$

Now, we re-organize ℓm 2-form as

$$\begin{pmatrix} Z_1^{\langle 1 \rangle} & Z_1^{\langle 2 \rangle} & \cdots & Z_1^{\langle \ell \rangle} \\ Z_2^{\langle 1 \rangle} & Z_2^{\langle 2 \rangle} & \cdots & Z_2^{\langle \ell \rangle} \\ \vdots & \vdots & \ddots & \vdots \\ Z_m^{\langle 1 \rangle} & Z_m^{\langle 2 \rangle} & \cdots & Z_m^{\langle \ell \rangle} \end{pmatrix}$$

then every 2-forms in the column will contribute a factor of $\frac{1}{1-tz_i^2}$ and $\frac{1}{1-tyz_iz_j}$ while every 2-forms in the row will contribute a factor of $\frac{1}{1-tyz_iz_j}$ only.

Finally, we build the cochain complex similar to Appendix B and under Assumption 1, we conclude that

$$\operatorname{rank} \varPhi_{SNOVA}^{\langle \ell m \rangle} = \sum_{r=1}^{\lfloor \frac{\ell o}{2} \rfloor} (-1)^{r+1} \sum_{\substack{0 \le a_i \le o \\ \sum_i a_i = 2r}} \left(\sum_{k=0}^{r} N_{r,k}(t, \mathbf{z}, y; \mathbf{a}) \right) \prod_{i=1}^{\ell} \binom{n}{v + a_i}$$

where

$$N_{r,k}(t, \mathbf{z}, y; \mathbf{a}) = \left[z_1^{a_1} \cdots z_l^{a_l}\, t^r y^k\right] \left(\prod_{i=1}^{\ell} \frac{1}{1 - t\, z_i^2} \prod_{1 \le i < j \le \ell} \frac{1}{(1 - t\, y\, z_i z_j)^2} \right)^m .$$

A toy example We take

$$n = 9, \quad v = 6, \quad m = 2, \quad \ell = 2, \qquad o = n - v = 3.$$

Thus there are two blocks, each with *capacity* at most $o = 3$. Let z_1, z_2 mark the occupancies of blocks $1, 2$, let t mark how many 2-forms are wedged, and let $\mathbf{v}$ mark the number of *splits* (cross-block contributions).

1. **Generating series.**
   Hence, for one row ($m = 1$):
   $$F_1(z_1, z_2; t, y) = \frac{1}{(1 - tz_1^2)(1 - tz_2^2)\,(1 - t\mathbf{v} z_1 z_2)^2}. \tag{15}$$
   For $m = 2$ rows:
   $$F(z_1, z_2; t, \mathbf{v}) = F_1(z_1, z_2; t, \mathbf{v})^2. \tag{16}$$
   Capacity ≤ 3 means: after expanding in z_1, z_2, keep only monomials $z_1^x z_2^y$ with $0 \le x, y \le 3$, i.e. apply the truncation operator
   $$[z_1^{\le 3} z_2^{\le 3}](\cdot).$$
2. **Coefficient extraction for $r = 1$: $[t^1 \mathbf{v}^k]$.**
   From (15) we have
   $$[t]\, F_1 \;=\; z_1^2 + z_2^2 + 2\mathbf{v}\, z_1 z_2,$$
   because $(1 - t\mathbf{v} z_1 z_2)^{-2} = 1 + 2(t\mathbf{v} z_1 z_2) + \cdots$. Using (16),
   $$[t]\, F \;=\; 2[t]F_1 \;=\; 2z_1^2 + 2z_2^2 + 4\mathbf{v}\, z_1 z_2.$$
   All monomials satisfy the capacity bound, so truncation does nothing. Thus the *counts* are
   $$k = 0: \;4 \quad (\text{from } 2z_1^2 + 2z_2^2), \qquad k = 1: \;4 \quad (\text{from } 4\mathbf{v} z_1 z_2).$$
3. **Coefficient extraction for $r = 2$: $[t^2 \mathbf{v}^k]$.**
   First,
   $$[t^2]\, F_1 = z_1^4 + z_2^4 + z_1^2 z_2^2 + 2\mathbf{v}(z_1^3 z_2 + z_1 z_2^3) + 3\mathbf{v}^2 z_1^2 z_2^2.$$
   Then
   $$[t^2]\, F = 2[t^2]F_1 + \big([t]F_1\big)^2 = 3z_1^4 + 3z_2^4 + 4z_1^2 z_2^2 + 8\mathbf{v}(z_1^3 z_2 + z_1 z_2^3) + 10\mathbf{v}^2 z_1^2 z_2^2.$$

Apply capacity ≤ 3: drop z_1^4 and z_2^4. The truncated part is

$$[z_1^{\leq 3} z_2^{\leq 3}]\,[t^2]F = 4z_1^2 z_2^2 + 8\mathbf{v}(z_1^3 z_2 + z_1 z_2^3) + 10\mathbf{v}^2 z_1^2 z_2^2.$$

So the counts are

$$k = 0:\ 4, \qquad k = 1:\ 16, \qquad k = 2:\ 10.$$

4. **Coefficient extraction for $r = 3$: $[t^3\mathbf{v}^k]$.**
Under capacity ≤ 3, the only feasible monomial at total occupancy $2r = 6$ is $z_1^3 z_2^3$. One computes

$$[z_1^{\leq 3} z_2^{\leq 3}]\,[t^3]F = (16\mathbf{v} + 20\mathbf{v}^3)\, z_1^3 z_2^3.$$

Hence the counts are

$$k = 1:\ 16, \qquad k = 3:\ 20.$$

5. **Assembling the rank (set $\mathbf{v} = 1$ and apply the weights).**
Here $n = 9$, $v = 6$, $o = 3$, so the weight polynomial is

$$B(z) = \sum_{x=0}^{3} \binom{9}{6+x} z^x = 84 + 36z + 9z^2 + z^3.$$

Let $\langle \cdot, \cdot \rangle$ denote coefficientwise inner product in z_1, z_2. Then the rank is

$$\text{rank} = \sum_{r=1}^{3} (-1)^{r+1}\,[t^r]\Big\langle [z_1^{\leq 3} z_2^{\leq 3}]\,F(z_1, z_2; t, \mathbf{v}{=}1),\ B(z_1)B(z_2) \Big\rangle.$$

Evaluating the three layers gives

$$r = 1:\ +8208, \qquad r = 2:\ -1710, \qquad r = 3:\ +36,$$

and therefore
$$\boxed{\text{rank} = 8208 - 1710 + 36 = 6534}$$
which matches the observed rank for $(n, v, m, \ell) = (9, 6, 2, 2)$ in Table 6.

D Cost of the attack in quadratic equations modeling

Table 4. Estimated parameters of the quadratic modeling when we eliminate variables. Note that we use balanced projection where **o'** are taken from Table 3, constant factors are not included, and the dash indicates that we found no **o'** that enables our attack.

Variants	v	m	ℓ	o'	$\log_2 E_0$	$\log_2 U$	$\log_2(U^\omega)$
SNOVA-I ($q = 16$)	37	17	2	8	27.5	48.6	136
	25	8	3	5	16.4	47.2	132
	24	5	4	5	16.0	62.6	176
SNOVA-III ($q = 16$)	56	25	2	11	40.2	71.4	201
	49	11	3	11	38.2	105.4	296
	37	8	4	7	24.7	93.2	262
	24	5	5	4	13.2	67.4	189
SNOVA-V ($q = 16$)	75	33	2	13	50.0	91.8	258
	66	15	3	14	50.3	138.9	390
	60	10	4	–	–	–	–
	29	6	5	5	17.3	84.0	236

E Experimental results for Proposition 2 and 3

Table 5. Actual rank of $\Phi^{\langle m\rangle}_{\mathrm{SNOVA}}$ when using m base 2-forms versus our estimation in Proposition 2. It should be noted that we only report the parameters where the number of linear independent equations is known to be smaller than the number of variables, so that we have a meaningful lower bound.

n	v	m	ℓ	Rank	Formula 9	Difference
			$\ell = 2$			
5	3	1	2	43	43	0
5	3	2	2	83	83	0
6	4	2	2	125	125	0
7	4	1	2	793	793	0
7	5	2	2	175	175	0
8	6	2	2	233	233	0
8	6	3	2	345	345	0
8	6	4	2	454	454	0
8	5	1	2	1498	1498	0
8	5	2	2	2700	2700	0
9	5	1	2	12075	12075	0
9	6	1	2	2576	2576	0
9	6	2	2	4773	4773	0
9	6	3	2	6598	6598	0
10	7	1	2	4137	4137	0
10	7	2	2	7802	7802	0
10	7	3	2	11002	11002	0
			$\ell = 3$			
3	2	2	3	18	18	0
4	3	2	3	24	24	0
4	3	3	3	36	36	0
4	3	4	3	48	48	0
4	3	5	3	60	60	0
7	5	1	3	4126	4126	0
7	5	2	3	7677	7677	0
8	6	2	3	13366	13366	0

Table 6. Actual rank of $\Phi_{\text{SNOVA}}^{\langle \ell m \rangle}$ when using ℓm scaled 2-forms versus our estimation in Proposition 3. It should be noted that we only report the parameters where the number of linear independent equations is known to be smaller than the number of variables, so that we have a meaningful lower bound.

n	v	m	ℓ	Rank	Formula 10	Difference
5	3	1	2	66	66	0
6	4	1	2	98	98	0
6	4	2	2	190	190	0
7	4	1	2	1098	1098	0
7	5	1	2	136	136	0
7	5	2	2	266	266	0
7	5	3	2	390	390	0
8	5	1	2	2102	2102	0
8	6	1	2	180	180	0
8	6	2	2	354	354	0
8	6	3	2	522	522	0
8	6	4	2	684	684	0
9	5	1	2	15489	15489	0
9	6	1	2	3642	3642	0
9	6	2	2	6534	6534	0
9	7	1	2	230	230	0
9	7	2	2	454	454	0
9	7	3	2	672	672	0
9	7	4	2	884	884	0
9	7	5	2	1090	1090	0
9	7	6	2	1290	1290	0
10	6	1	2	35331	35331	0
10	7	1	2	5876	5876	0
10	7	2	2	10816	10816	0
10	8	1	2	286	286	0
10	8	2	2	566	566	0
10	8	3	2	840	840	0
10	8	4	2	1108	1108	0
10	8	5	2	1370	1370	0
10	8	6	2	1626	1626	0
10	8	7	2	1876	1876	0
11	8	1	2	8982	8982	0
11	8	2	2	16822	16822	0
11	8	3	2	23540	23540	0
3	2	1	3	18	18	0
4	3	1	3	24	24	0
4	3	2	3	48	48	0
5	4	1	3	30	30	0
5	4	2	3	60	60	0
5	4	3	3	90	90	0
5	4	4	3	120	120	0
6	4	1	3	3105	3105	0
6	5	1	3	36	36	0
6	5	2	3	72	72	0
6	5	3	3	108	108	0
6	5	4	3	144	144	0
6	5	5	3	180	180	0
7	5	1	3	6381	6381	0